M365

Fresh Tenant Setup

These are the steps I typically take to set up a fresh M365 E5 tenant.

Work in Progress

This is very much a continuous work in progress. I publish changes as I go. Screenshots might be out of date.

Use at your own risk

These are my personal steps. This should not be construed as official guidance. Always refer to the official Microsoft documentation available at learn.microsoft.com

Entra

Entra Cloud Sync

🔗 Portal 📘 Docs

Cloud Sync is the lightweight replacement for AAD Connect. Follow the instructions in the Docs link for a step-by-step example for a single forest install.

For conditional access, be sure to exclude the Directory Synchronization Accounts role from any MFA policies.

Hybrid Cloud Trust

📘 Docs

Create EntraID Kerberos Server

```
# Install the AzureADHybridAuthenticationManagement module
Install-Module -Name AzureADHybridAuthenticationManagement -AllowClobber

# Specify the on-premises Active Directory domain. A new Azure AD
# Kerberos Server object will be created in this Active Directory domain.
$domain = "contoso.com"
```

```
# Enter a UPN of an Azure Active Directory global administrator
$userPrincipalName = "admin@contoso.com"

# Enter a domain administrator username and password.
$domainCred = Get-Credential

# Create the new Azure AD Kerberos Server object in Active Directory
# and then publish it to Azure Active Directory.
# Open an interactive sign-in prompt with given username to access the Azure
AD.
Set-AzureADKerberosServer -Domain $domain -UserPrincipalName
$userPrincipalName -DomainCredential $domainCred

# Verify server
Get-AzureADKerberosServer -Domain $domain -DomainCredential $domainCred -
UserPrincipalName $userPrincipalName

Id                  : 17530
UserAccount         : CN=krbtgt_AzureAD,CN=Users,DC=contoso,DC=com
ComputerAccount     : CN=AzureADKerberos,OU=Domain
Controllers,DC=contoso,DC=com
DisplayName         : krbtgt_17530
DomainDnsName       : contoso.com
KeyVersion          : 27591
KeyUpdatedOn        : 10/13/2022 9:23:43 PM
KeyUpdatedFrom      : CONTOSO-DC-01.contoso.com
CloudDisplayName    : krbtgt_17530
CloudDomainDnsName  : contoso.com
CloudId             : 17530
CloudKeyVersion     : 27591
CloudKeyUpdatedOn   : 10/13/2022 9:23:43 PM
CloudTrustDisplay   :
```

Device Settings

🔗 Portal

- Disable adding GA to local admin.

Microsoft Entra join and registration settings

Users may join devices to Microsoft Entra

All | Selected | None

Selected

No member selected

Users may register their devices with Microsoft Entra

All | None

Learn more on how this setting works

Require Multifactor Authentication to register or join devices with Microsoft Entra

Yes | No

We recommend that you require Multifactor Authentication to register or join devices with Microsoft Entra using Conditional Access. Set this device setting to No if you require Multifactor Authentication using Conditional Access.

Maximum number of devices per user

50

Local administrator settings

Global administrator role is added as local administrator on the device during Microsoft Entra join (Preview)

Yes | No

Registering user is added as local administrator on the device during Microsoft Entra join (Preview)

All | Selected | None

Selected

No member selected

Manage Additional local administrators on all Microsoft Entra joined devices

Enable Microsoft Entra Local Administrator Password Solution (LAPS)

Yes | No

Other settings

Restrict users from recovering the BitLocker key(s) for their owned devices

Yes | No

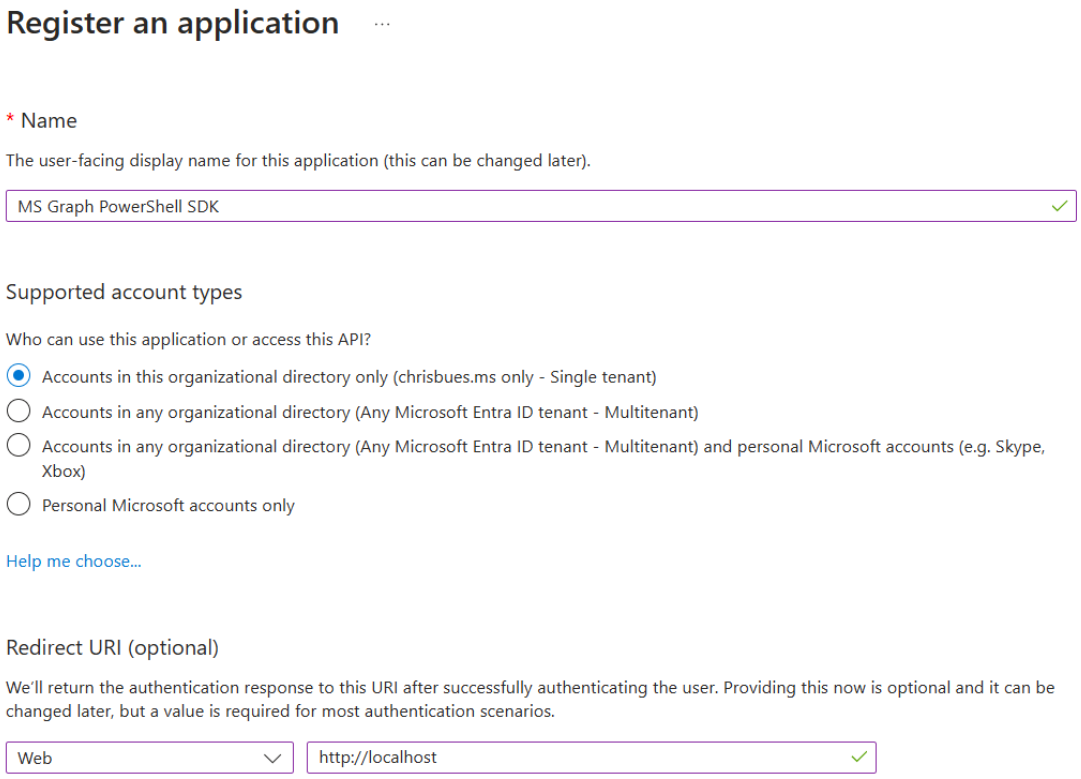
App Registrations

MS Graph PowerShell SDK

To enable use of the MS Graph PowerShell SDK, create an app registration for app-only for use with the SDK.

🔗 Portal

1. Create the app registration.



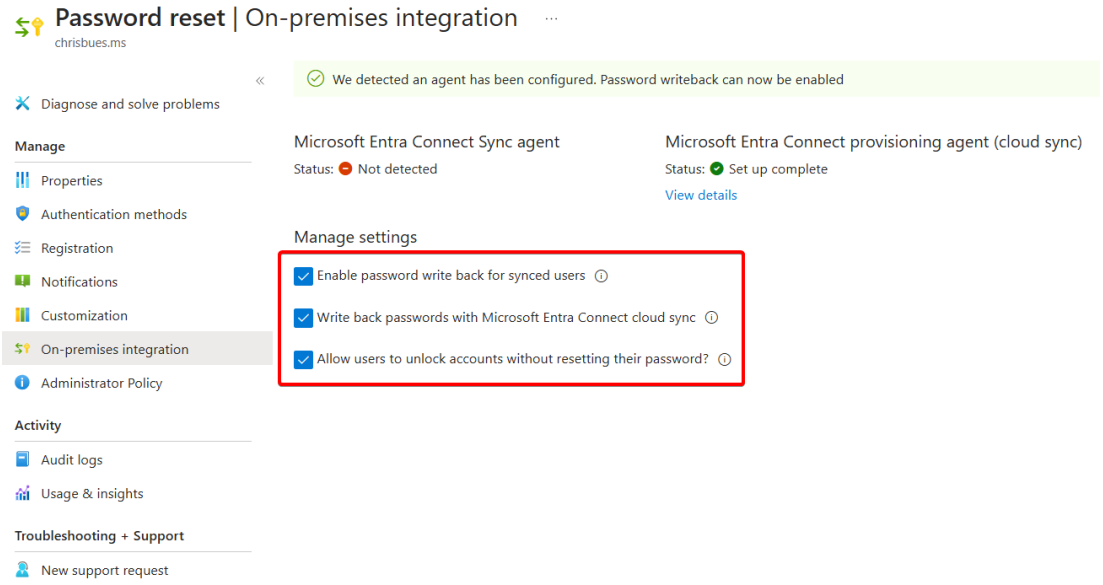
2. Grant application permissions for Microsoft Graph that are necessary for your use cases.
3. I prefer to do app consent cert-based authentication. Here's a couple of links explaining the process:
   a. How to use Connect-MgGraph - All Options — LazyAdmin
   b. Use app-only authentication with the Microsoft Graph PowerShell SDK | Microsoft Learn

Identity Protection

Self Service Password Reset

🔗 Portal 📘 Docs

- Enable Self service password reset
- Target a group that excludes service accounts. Easiest way to do this is with a dynamic group. Example rule: `(user.displayName -ne "On-Premises Directory Synchronization Service Account") and (user.userPrincipalName -notStartsWith "svc")`
- Enable Password writeback in On-premises integration



Authentication Methods

🔗 Portal 📘 Docs

POLICIES

For methods, enable

- Passkeys

- Authenticator

Enable and Target **Configure**

Note: Users must be included as part of the Microsoft Authenticator targeted groups under the 'Enable and Target' tab.

GENERAL

Allow use of Microsoft Authenticator OTP Yes No

Require number matching for push notifications

Note: This feature has been enabled for all users of the Microsoft Authenticator. Learn more

Status Enabled

Target **Include**

All users

Select group

Show application name in push and passwordless notifications

Note: If the feature status is set to Microsoft-managed, it will be enabled by Microsoft at an appropriate time after the preview. Learn more

Status Enabled

Target **Include** Exclude

All users

Select group

Show geographic location in push and passwordless notifications

Note: If the feature status is set to Microsoft-managed, it will be enabled by Microsoft at an appropriate time after the preview. Learn more

Status Enabled

Target **Include** Exclude

All users

Select group

Microsoft Authenticator on companion applications

Note: If the feature status is set to Microsoft-managed, it will be enabled by Microsoft at an appropriate time after the preview. Learn more

Status Disabled

Target **Include** Exclude

All users

Select group

- Enable FIDO2, Authenticator, Temporary Access Pass in Authentication Methods

AUTHENTICATION METHODS MIGRATION

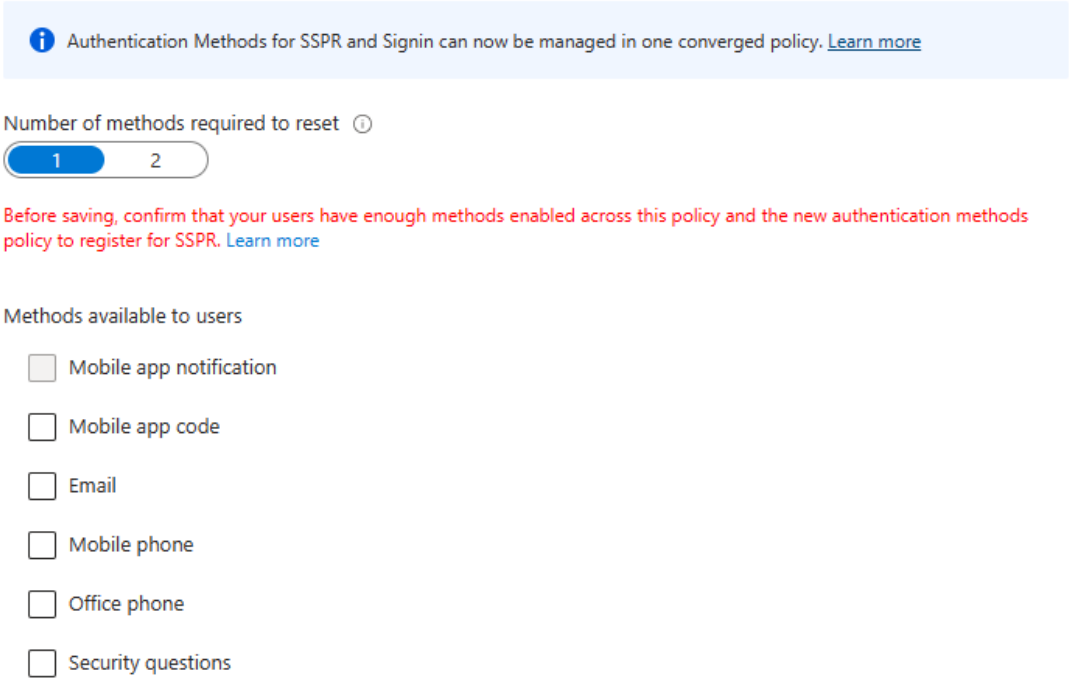
- Disable verification options in the legacy MFA settings portal

Methods available to users:

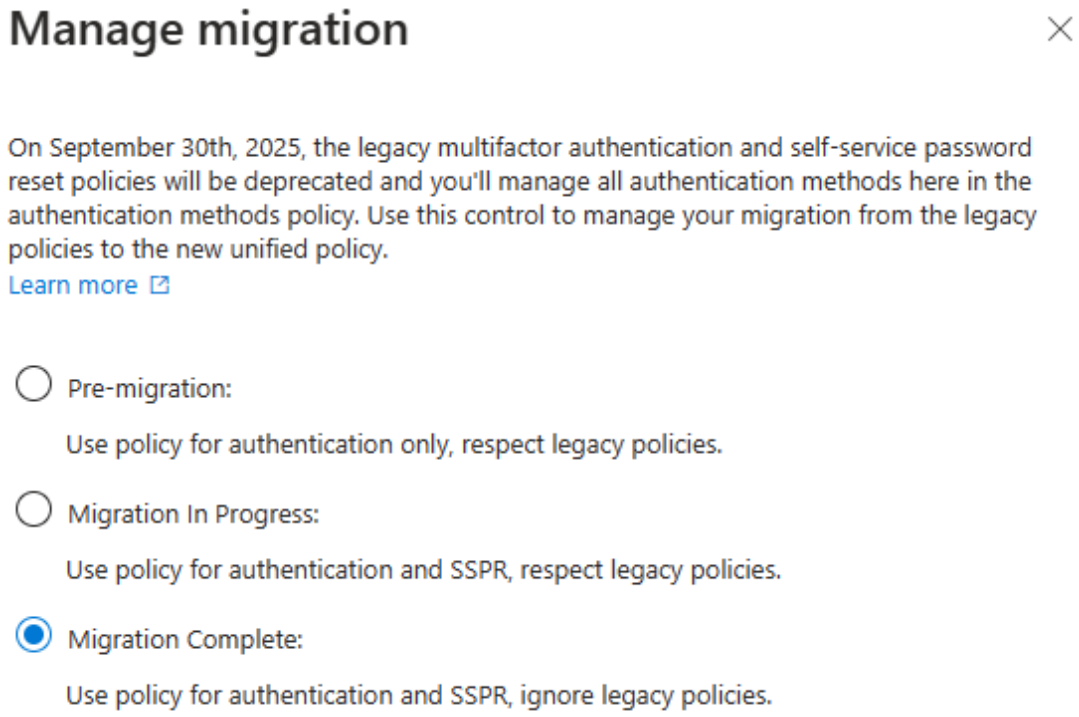
- [ ] Call to phone
- [ ] Text message to phone
- [ ] Notification through mobile app
- [ ] Verification code from mobile app or hardware token

Disabling all authentication methods could lock out your users. Ensure that you have enough authentication methods enabled in the new authentication methods policy before saving. Learn more

- Disable Authentication methods in SSPR Authentication Methods



- Migrate to the Converged Authentication Methods Policy

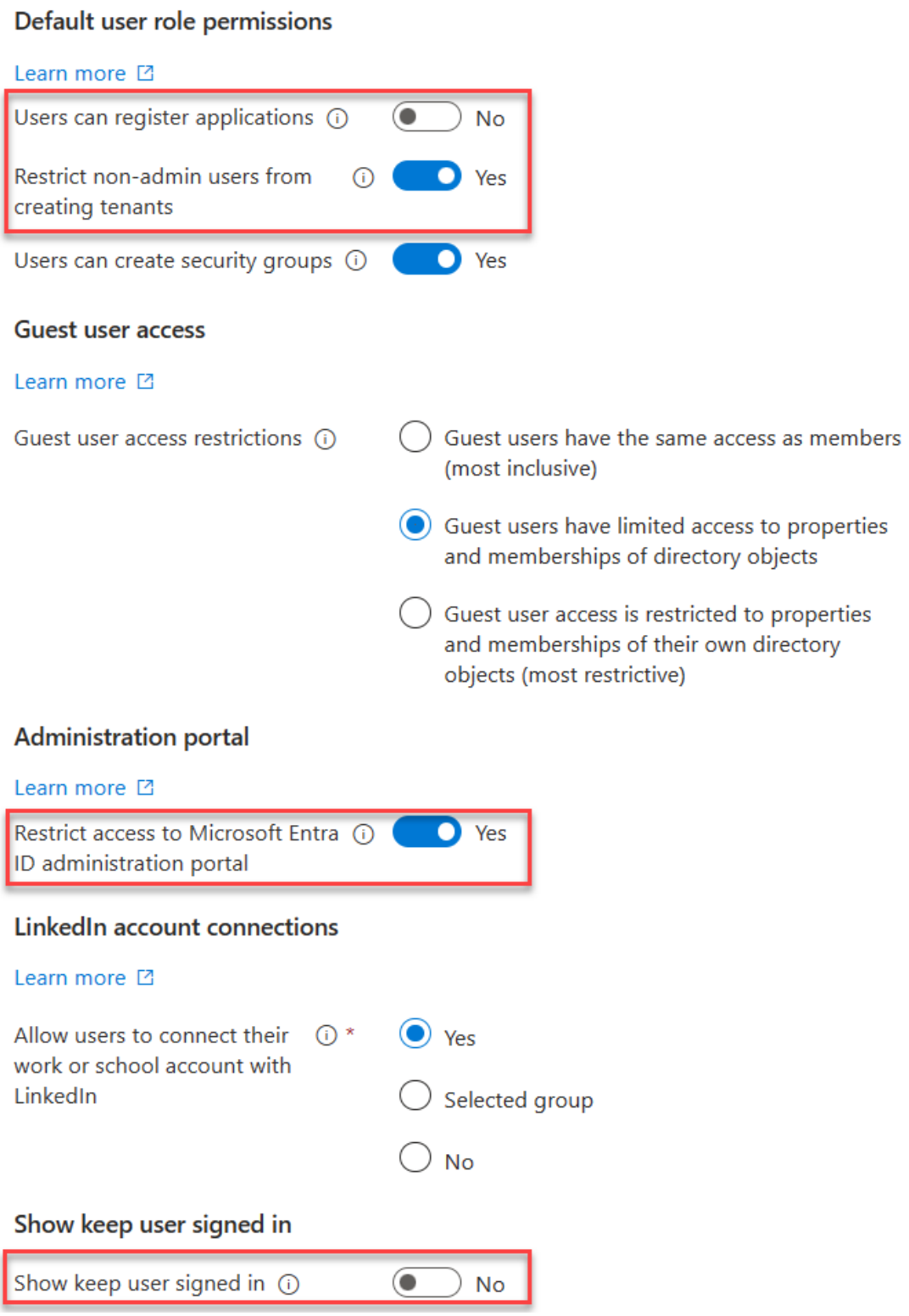


User Settings

Portal

- Toggle Off

- Users can register Applications
- Show keep user signed in
- Toggle On
- Restrict non-admin users from creating tenants
- Restrict access to Entra ID administration portal



User Feature Settings

Portal

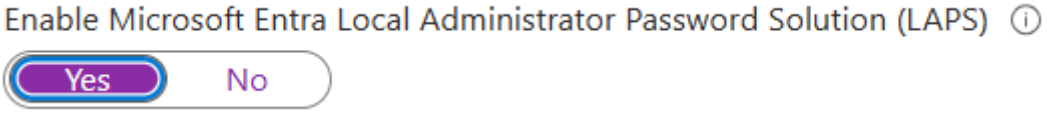
- Select All for Users can use preview features for My Apps

Device Settings

Cloud LAPS

Portal Docs

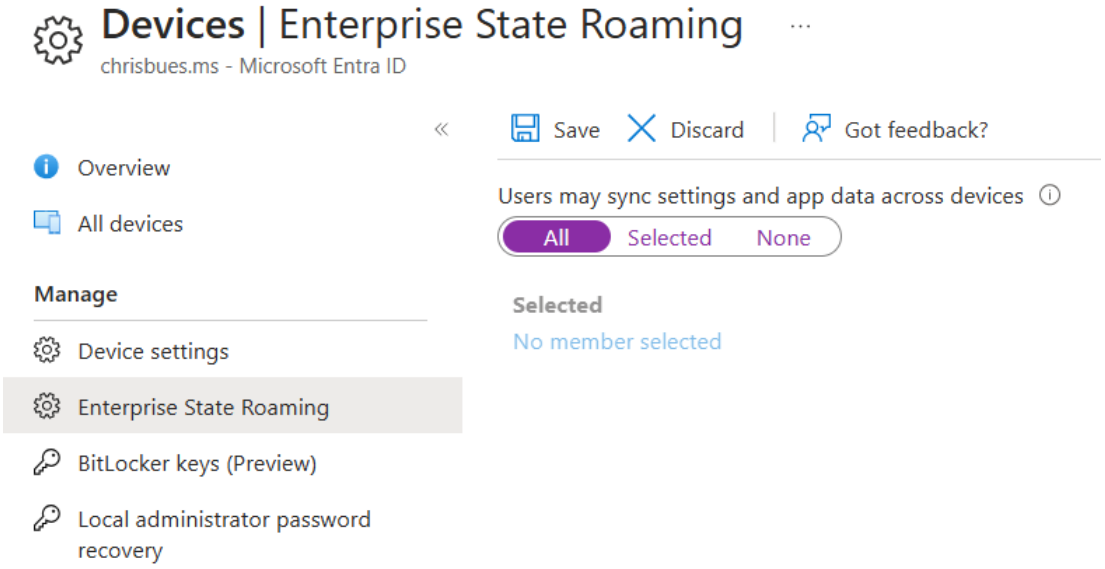
- Enable LAPS



Enterprise State Roaming

Portal

- Enable Enterprise State Roaming

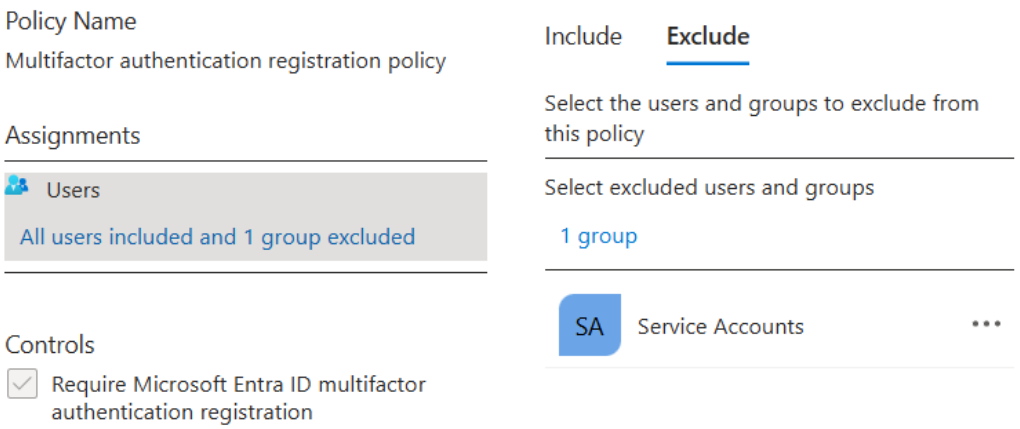


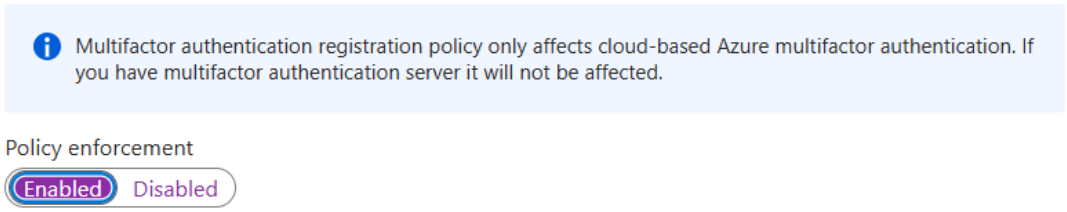
Identity Protection

Multifactor authentication registration policy

- Create a EntraID group called Service Accounts, add the Entra Cloud sync account

- Enable the policy, targeting all users and excluding the group you just created.



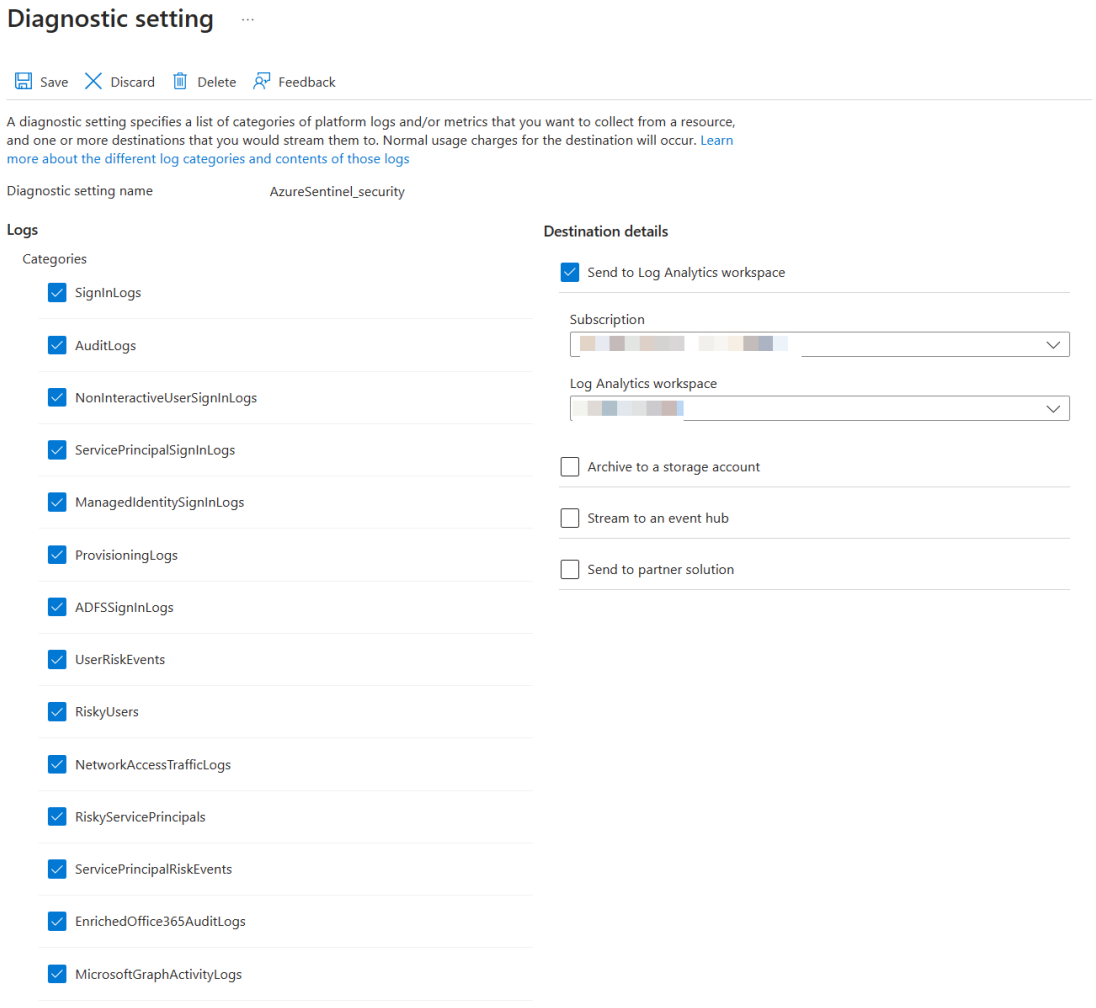


Diagnostic Settings

Portal Docs

Prior to doing so, create a Log Analytics workspace and add Sentinel to it.

- Enable all diagnostic settings to log to your Sentinel's log analytics workspace



Global Secure Access

🔗 Portal 📘 Docs

Enable GSA

Click Activate to enable GSA in your tenant

Home > Authentication methods | Policies > chrisbues.ms >

Welcome to Global Secure Access ...

Global Secure Access is now generally available. Licensing requirements have been updated. Learn more

Activate the Global Secure Access, which includes Microsoft Entra Internet Access and Microsoft Entra Private Access. Learn more

1. Global Secure Access prerequisites

- You have the required administrator role to activate the Global Secure Access.
- You have the required license to start using Global Secure Access

A Global Secure Access Administrator, Security Administrator, or Global Administrator must be assigned to activate and manage Global Secure Access features Learn more

2. Activate Global Secure Access in your tenant

To activate Global Secure Access, click the activate button below. Note activation will not impact any workload in your tenant until preview features are enabled.

Activate

3. Get started with Global Secure Access

Learn about Global Secure Access and the next steps for getting started.

Get Started

Internet Access

1. Enable the Microsoft Profile

Home > Get started > chrisbues.ms >

Traffic forwarding ...

Global Secure Access is now generally available. Licensing requirements have been updated. Learn more

Enabling this profile directs Global Secure Access clients to acquire traffic for this profile.

Are you sure you want to proceed?

OK Cancel

Traffic forwarding profiles enable admins to forward specific traffic to Global Secure Access. Assign traffic forwarding profiles to users running the Global Secure Access client. For clientless devices, use the Microsoft profile to assign to remote networks. Learn more

Microsoft traffic profile	Private access profile	Internet access profile
Disabled	Disabled	Disabled
Last modified on 08/15/2024, 11:42 PM	*Last modified on 08/15/2024, 11:42 PM*	*Last modified on not available*
Applies to Internet traffic to Microsoft services	Applies to Private resources	Applies to All internet traffic, except for the Microsoft traffic profile
Microsoft traffic policies 3 policies View	Private access policies Quick Access, 0 Applications	Internet access policies 3 policies View
Linked Conditional Access policies None	Linked Conditional Access policies None	Linked Conditional Access policies None
User and group assignments 0 users, 0 groups assigned View	User and group assignments 0 users, 0 groups assigned View	User and group assignments 0 users, 0 groups assigned View
Remote network assignments 0 assigned remote networks View	Remote network assignments Not applicable	Remote network assignments Not applicable

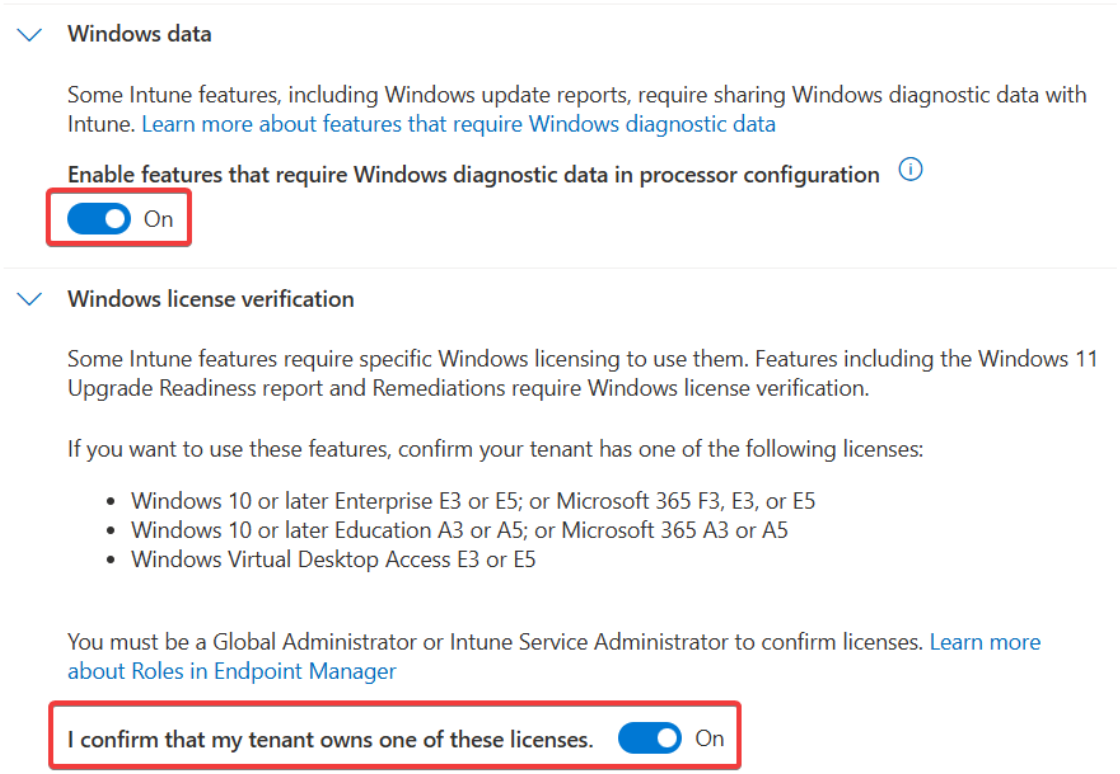
2. Download the GSA Client and deploy to Windows devices.

Intune

Tenant Administration Settings

Windows Data Connector

Portal



Defender for Endpoint Connector

Portal

You need to enable the Defender side first.

Endpoint Security Profile Settings

Allow Microsoft Defender for Endpoint to enforce Endpoint Security Configurations	Off **On**

Compliance policy evaluation

Connect Android devices version 6.0.0 and above to Microsoft Defender for Endpoint	Off **On**
Connect iOS/iPadOS devices version 13.0 and above to Microsoft Defender for Endpoint	Off **On**
Connect Windows devices version 10.0.15063 and above to Microsoft Defender for Endpoint	Off **On**
Enable App Sync (sending application inventory) for iOS/iPadOS devices	Off **On**
Send full application inventory data on personally owned iOS/iPadOS devices	**Off** On
Block unsupported OS versions	**Off** On

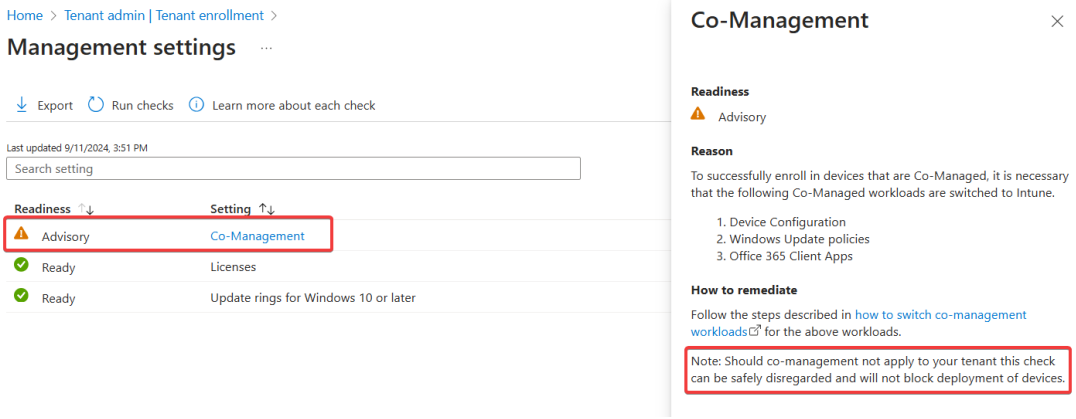
App protection policy evaluation

Connect Android devices to Microsoft Defender for Endpoint	Off **On**
Connect iOS/iPadOS devices to Microsoft Defender for Endpoint	Off **On**

Windows Autopatch

🔗 Portal 📘 Docs

- Run the prereq check. You'll see an advisory for co-management, this can be safely disregarded.



- Grant admin access for Microsoft

Allow administrator access for Microsoft

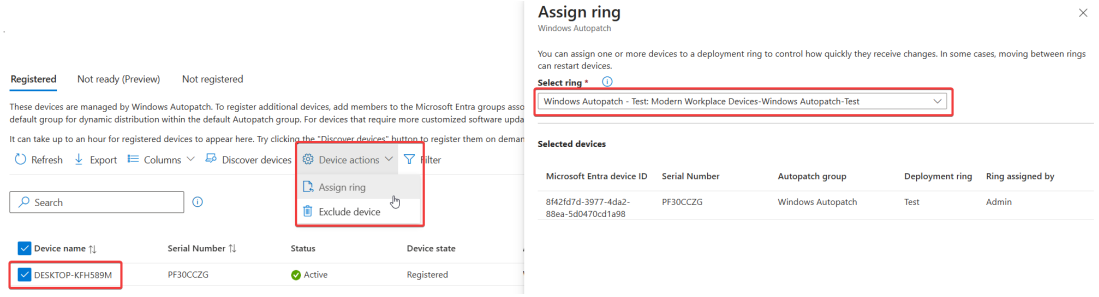
To get started, Microsoft needs your permission to take a few actions in your Microsoft Entra organization and on devices you want to enroll in Windows Autopatch. With your permission, Microsoft will do the following:

- Create a Microsoft application that we use to run the Windows Autopatch service. Learn more about Windows Autopatch enterprise applications
- Create the policies, groups and scripts necessary to run the service. This involves excluding Windows Autopatch device groups, where applicable, for any of your existing policies that may cause conflicts. Windows Autopatch update policies must take precedence to avoid any conflicts. Learn more about Changes made at tenant enrollment
- Manage devices using Intune.
- Collect and share info on usage, status, and compliance for devices and apps.
- Collect and share Windows Diagnostic data on usage, status, and compliance for devices and apps. Learn more about the data we collect
- Store Windows Autopatch data securely in Azure data centers based on your data residency. Learn more about Windows Autopatch data storage

☑ I give Microsoft permission to manage my Microsoft Entra organization on my behalf.
Revoking this access at any point terminates the service.

Agree

- Provide Admin contact info.
- Add devices to the the default autopatch group `Windows Autopatch Device Registration`
- Wait a few minutes, then ensure the devices show in the Windows Autopatch devices here
- Put a couple of devices in the Test ring by clicking on the device name, then selecting Device Actions -> Assign Ring. In the flyout, choose the Test ring

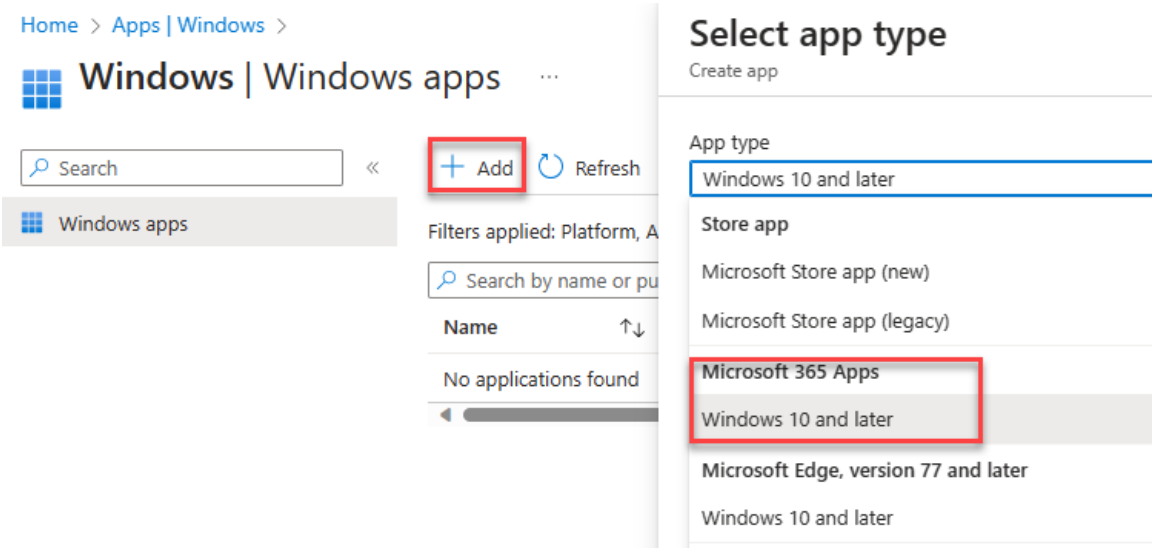


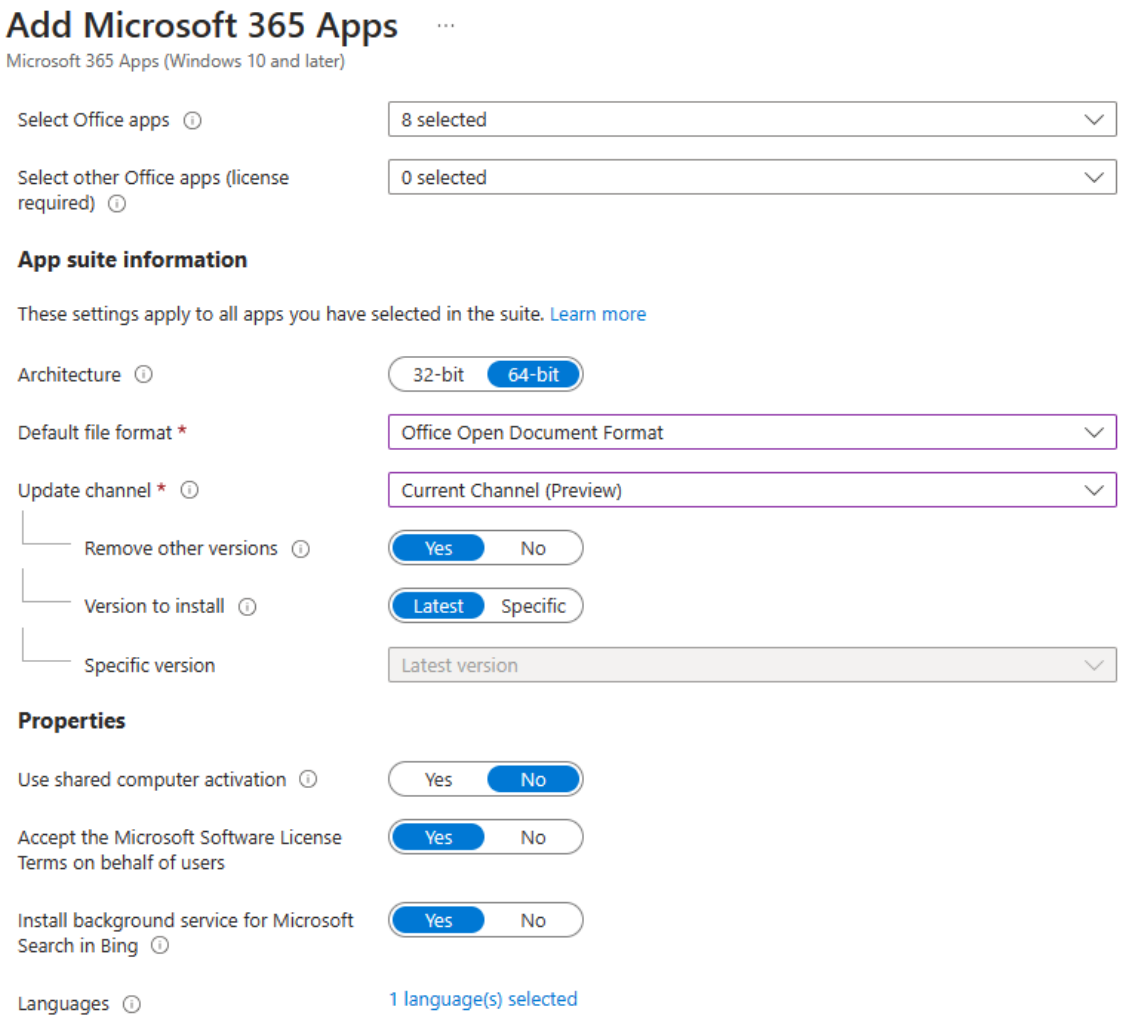
Applications

🔗 Portal 📘 Docs

Windows

Add app -> Microsoft 365 Apps for Windows 10 and Later. Assign to all devices.



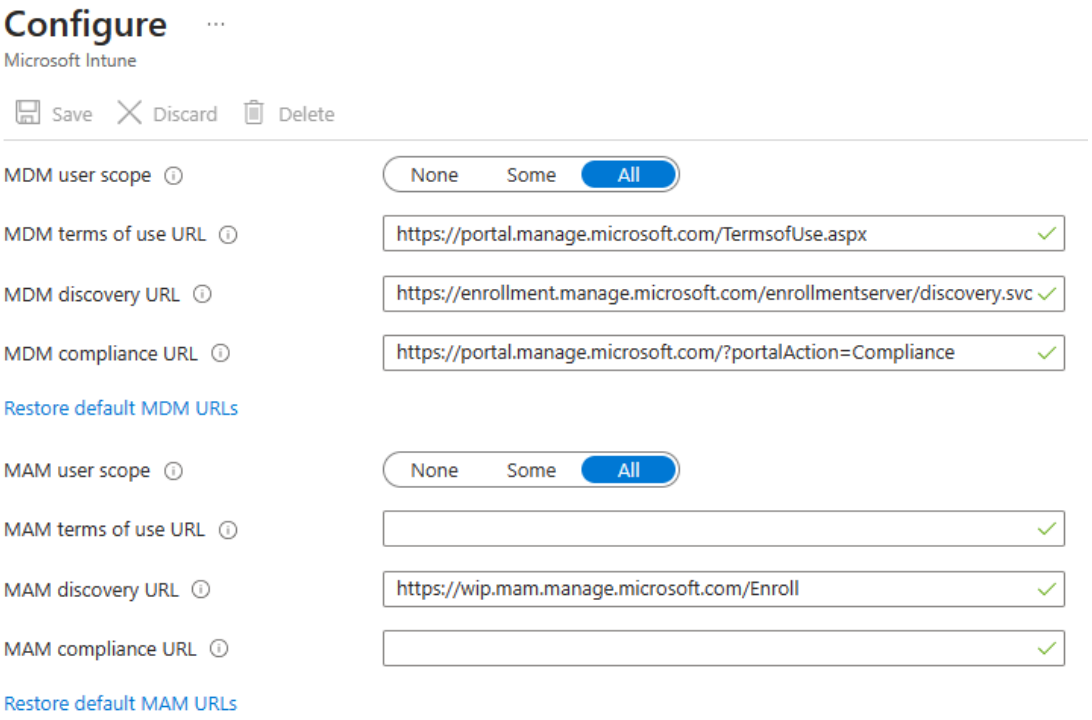


Devices

Windows Automatic Enrollment

🔗 Portal

- Set MDM and MAM user scopes to all



Windows Autopilot

🔗 Portal

Follow this guide: Overview for Windows Autopilot device preparation user-driven Microsoft Entra join in Intune | Microsoft Learn

Prereqs:

- Create Entra Groups
- Automatic Enrolment Set
- Enrolled devices. Windows Autopilot device preparation devices
  - Set App ID f1346770-5b25-470b-88bd-d5744ab7952c as the owner.
  - Targeted Users - Windows Autopilot device preparation users
- Create/update an Office deployment, target the device group created above.

Create a device prep policy

Device Enrollment

Windows Autopilot device preparation policies

Deployment settings | Review + save

Deployment settings

Deployment mode	User-driven *
Deployment type	Single user *
Join type	Microsoft Entra joined *
User account type	Administrator

Out-of-box experience settings

Minutes allowed before showing installation error	30
Custom error message	Contact your organization's support person for help. *
Allow users to skip setup after multiple attempts	Yes
Show link to diagnostics	Yes

Apps

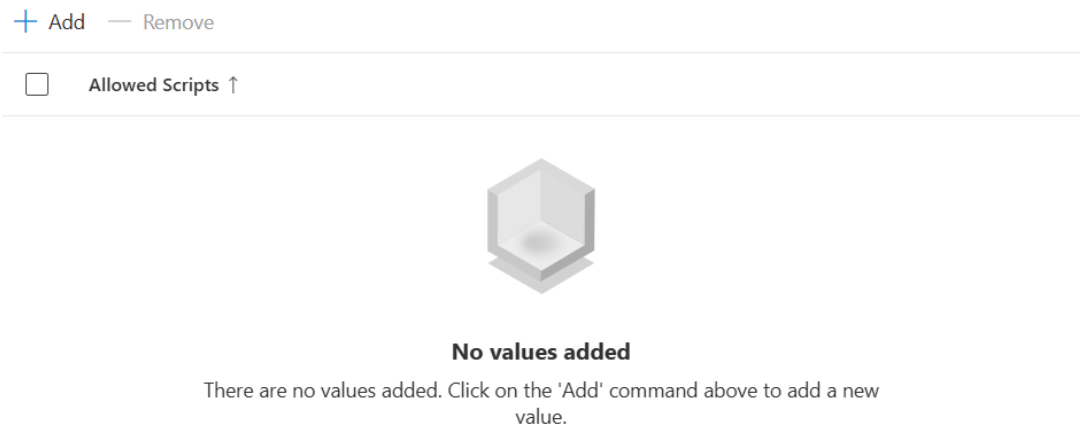
Select up to 10 managed apps you want to reference with this deployment. These apps should be assigned to the device security group you selected earlier. You can check the installation status for these apps in the device details for devices in this deployment.

+ Add — Remove

Allowed Applications ↑	Publisher	Version
Company Portal	Microsoft Corporation	
Microsoft 365 Apps for Windows 10 a...	Microsoft	

Scripts

Select up to 10 PowerShell scripts to install during this deployment. These scripts should be already assigned to the Device group selected earlier. You can check the installation status for these scripts in the device details for devices in this deployment.



iOS Enrollment

🔗 Portal

User-driven iOS enrollment is a two step process - the push certificate and the enrollment profile.

CONFIGURE APPLE MDM PUSH CERTIFICATE

📘 Docs

Configure MDM Push Certificate

Delete

Essentials

Status
Active

Days until expiration
365

Last updated
10/11/2023

Expiration
10/10/2024

Apple ID

Subject ID
com.apple.mgmt.External.

Serial number

You need an Apple MDM push certificate to manage Apple devices with Intune.
Steps:

1. I grant Microsoft permission to send both user and device information to Apple. More information on Microsoft permission.

   I agree.

2. Download the Intune certificate signing request required to create an Apple MDM push certificate.

   Download your CSR

3. Create an Apple MDM push certificate. More information on Apple MDM push certificate.

   Create your MDM push Certificate

4. Enter the Apple ID used to create your Apple MDM push certificate.

   Apple ID *

5. Browse to your Apple MDM push certificate to upload

   Apple MDM push certificate *
   "MDM_ Microsoft Corporation_Certificate.pem"

Upload

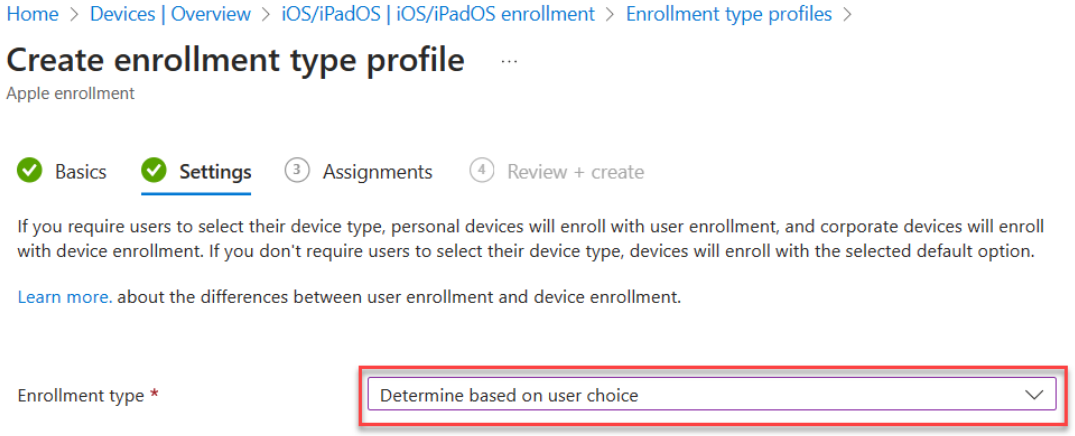
ENROLLMENT PROFILE

Portal

Native iOS enrollment

There's this nifty-keen account driven user enrollment available in iOS 15+, but you'll need a web server to serve up the json file Apple expects.

- Configure an enrollment profile
- Create a profile that allows user choice of type of device (corporate vs user), target all users.

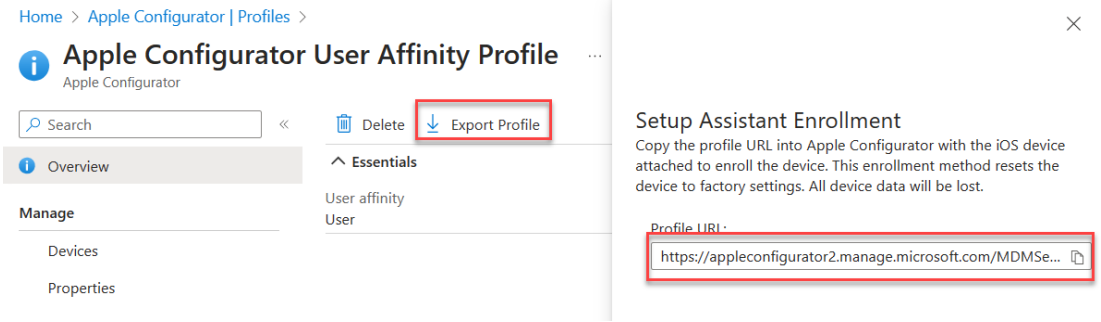


SUPERVISED IOS ENROLLMENT WITH APPLE CONFIGURATOR

🔗 Portal 📘 Docs

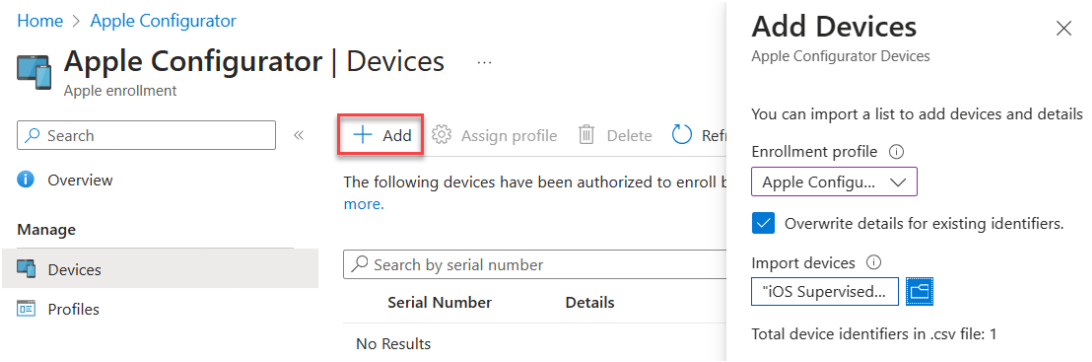
There are two options for Apple Config profile - with user affinity and without. For testing, enrollment with User Affinity with the Company Portal app mimics how devices might be distributed to end users.

1. Create a new Enrollment Profile. On the settings step, select: User affinity: `Enroll with User Affinity` Select where users must authenticate: `Company Portal`
2. Export the profile you just created. Copy the URL.

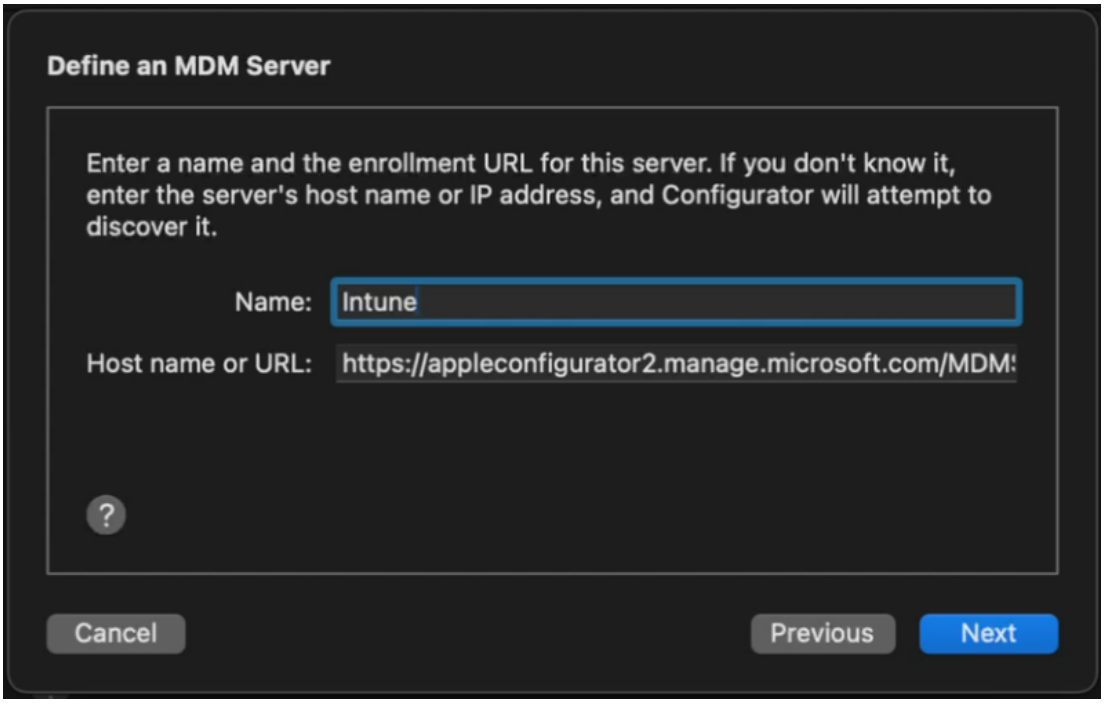


3. Create a csv file with the serial numbers of iPads you wish to enroll. `Serial number,device details`

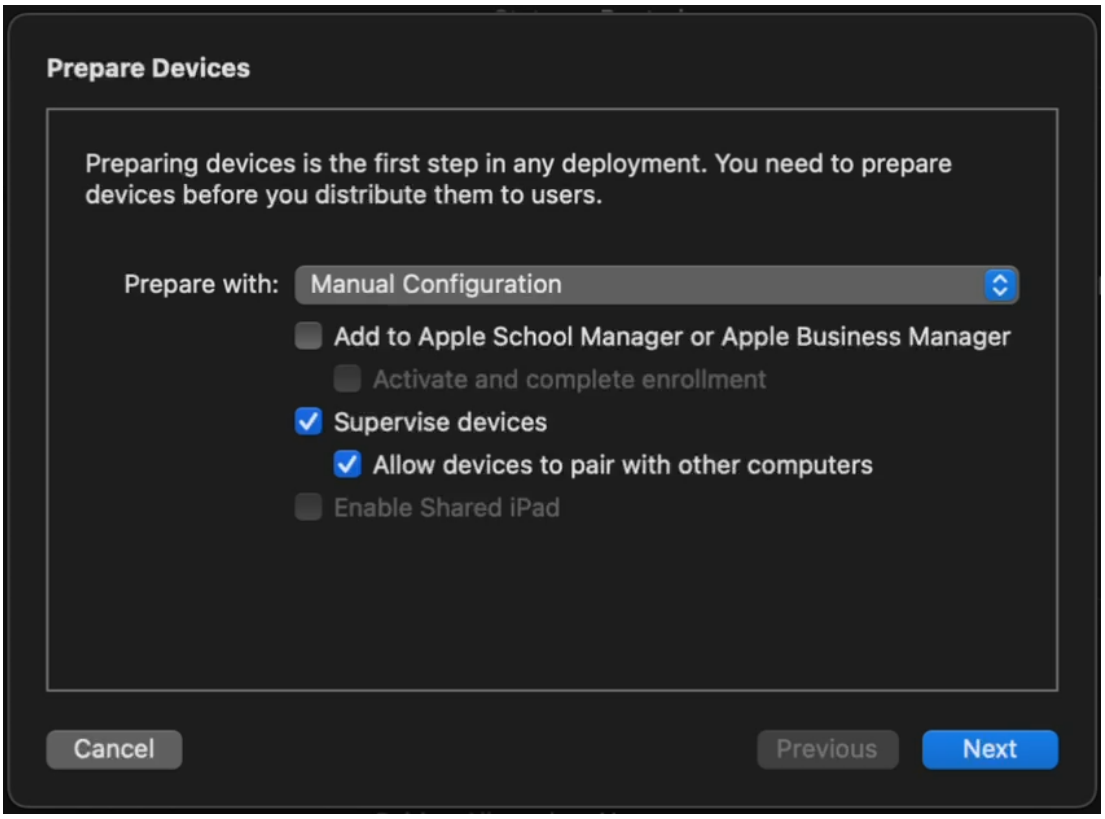
4. Upload the csv file in the portal under Devices. Assign the profile you just created.



5. In Apple Configurator, choose Settings -> Servers. Click + to add a server. Add the URL you copied from step 2.



6. Connect a device, and at the main screen, click Prepare. Leave the default options unchanged.



7. Choose the Intune MDM server defined in Step 5.

8. Skip Apple Business Manager sign-in if prompted. At the Organization screen select a previous org or create a new one. This is shown in the settings app in iOS.
9. Choose to generate a new supervision Identity or reuse an existing one.

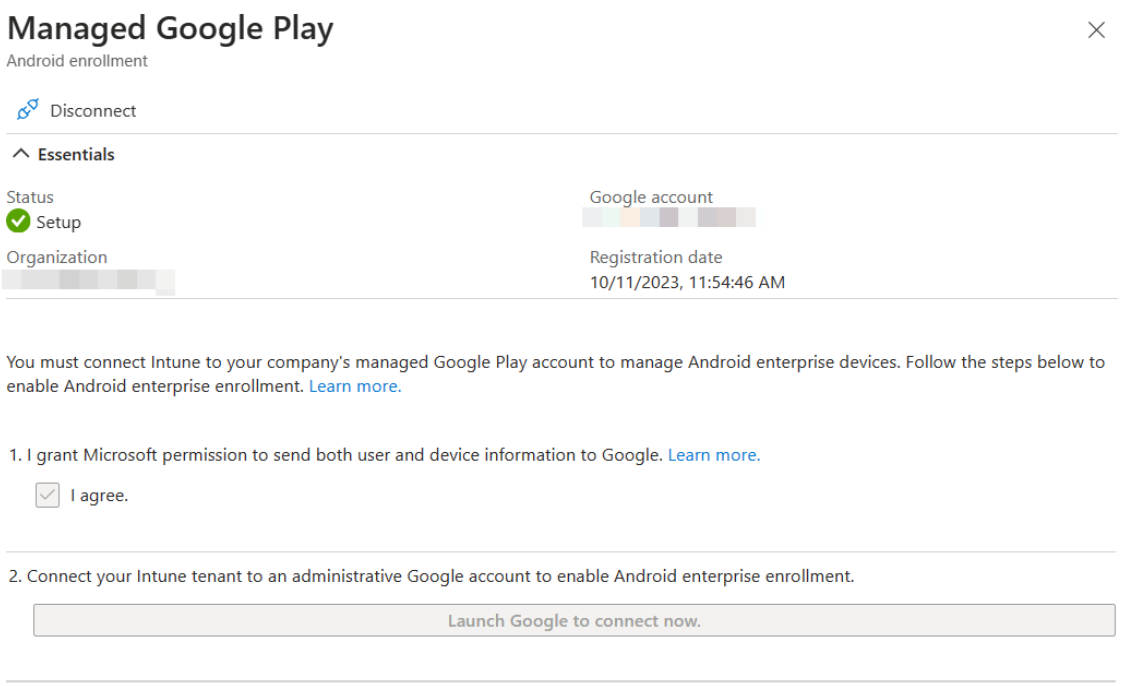
10. Choose which steps to display in the Setup Assistant. Click Prepare to start the process.

Android Enrollment

Portal Docs

User-driven Android enrollment is a two step process - the managed Google Play account linking and the enrollment profile.

MANAGED GOOGLE PLAY ACCOUNT LINKING



MacOS Enrollment

Portal Docs

As with MDE for MacOS, this tends to change, so be sure to check the docs for the most recent steps.

:material-**head**-sync: tl;dr

- Create a MacOS enrollment profile here if you didn't for iOS yet - they're shared between iOS and MacOS.
- Download the Company Portal app for MacOS from here and deploy the company portal app as a MacOS LOB app

App information Edit

Name	Company Portal
Description	CompanyPortal-Installer.pkg
Publisher	Microsoft
Category	No Category
Information URL	No Information URL
Privacy URL	No Privacy URL
Developer	No Developer
Owner	No Owner
Notes	No Notes
Logo	No logo

Program Edit

Pre-install script	No Pre-install script
Post-install script	No Post-install script

Requirements Edit

Minimum operating system	macOS Sonoma 14.0

Detection rules Edit

Ignore app version	Yes
Included apps	org.cocoapods.RxSwiftExt 6.2.1 org.cocoapods.AppleClientLogger 0.1.0 org.cocoapods.PowerLiftKit 5.2.0 org.cocoapods.Cache 5.2.0 com.microsoft.CPCore-PushNotifications-Mac 1.0 org.cocoapods.RxRelay 6.6.0 org.cocoapods.CocoaLumberjack 3.6.1 org.cocoapods.FluentIcons 1.1.136 org.cocoapods.FluentUI 0.2.10 org.cocoapods.Zip 2.1.1 org.cocoapods.MSAL 1.4.1 org.cocoapods.Alamofire 4.9.1 com.microsoft.AADNGCAuthenticationMacOS 1.0 com.microsoft.NGCKeyProviderMacOS 1.0 com.microsoft.MSAuthNetworkingMacOS 1.0.0 org.cocoapods.RxCocoa 6.6.0 com.microsoft.CommonFramework 1.0 org.cocoapods.RxSwift 6.6.0 com.microsoft.CompanyPortalMac 5.2409.1 com.microsoft.autoupdate2 4.74

Assignments Edit

Group mode	Group
Required	
Included	All devices
Available for enrolled devices	

MacOS Platform SSO

Use the instructions here as a guide.

tl;dr Use the settings below in a config profile to deploy platform sso with the following options:

- Password authentication which syncs the Entra password with the local account password
- Create new users as admins

Authentication

Remove category

Extensible Single Sign On (SSO)

Remove subcategory

Configure an app extension that enables single sign-on (SSO) for devices.

32 of 47 settings in this subcategory are not configured

Authentication Method (Deprecated): Password

Extension Identifier: com.microsoft.CompanyPortalMac.ssoextension

Platform SSO

Authentication Method: Password

Enable Create User At Login: Enabled

New User Authorization Mode: Admin

Token To User Mapping

Account Name: preferred_username

Full Name: name

Use Shared Device Keys: Enabled

Registration Token: {{DEVICEREGISTRATION}}

Screen Locked Behavior: Do Not Handle

Team Identifier: UBF8T346G9

Type *: Redirect

URLs

Delete Sort Import Export

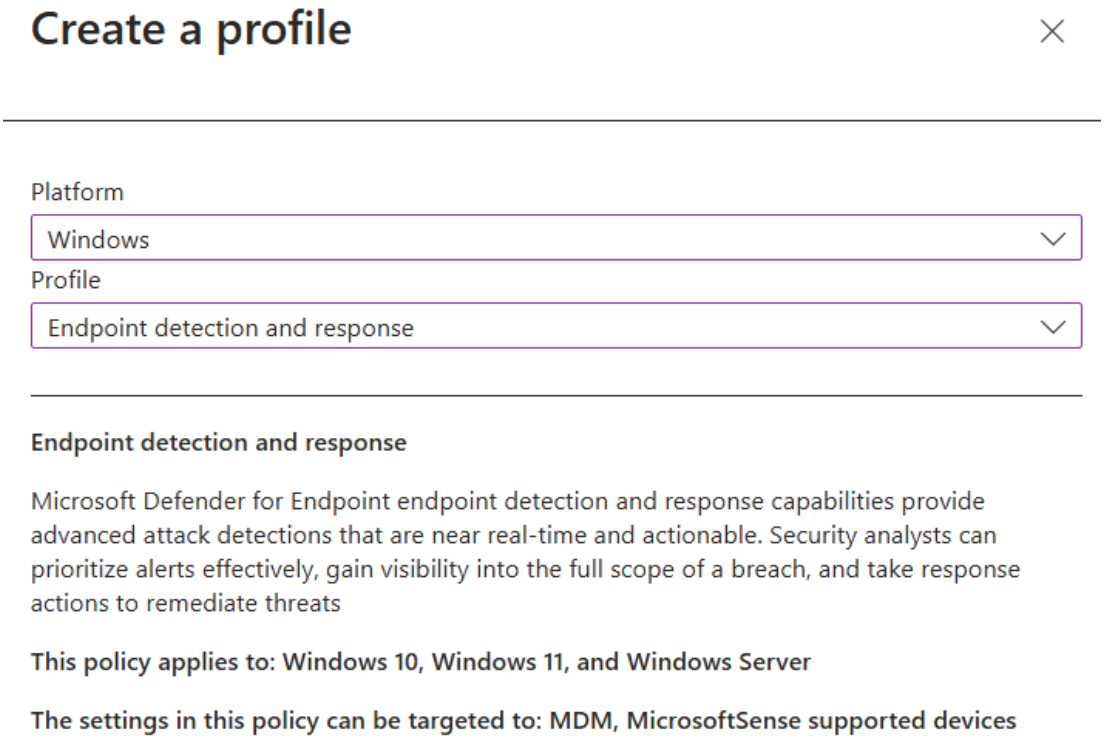
- https://login.microsoftonline.com
- https://login.microsoft.com
- https://sts.windows.net

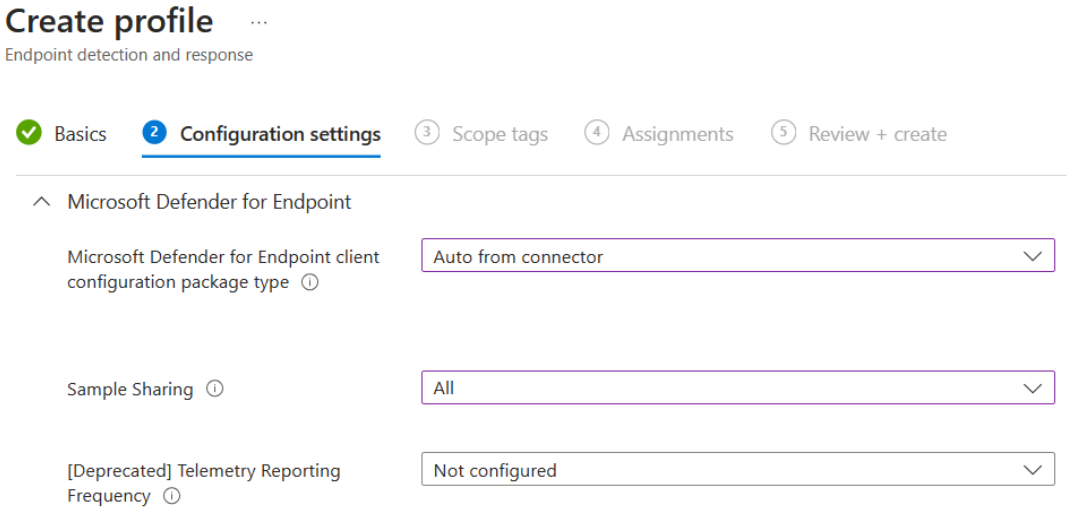
Endpoint security

Portal

Windows

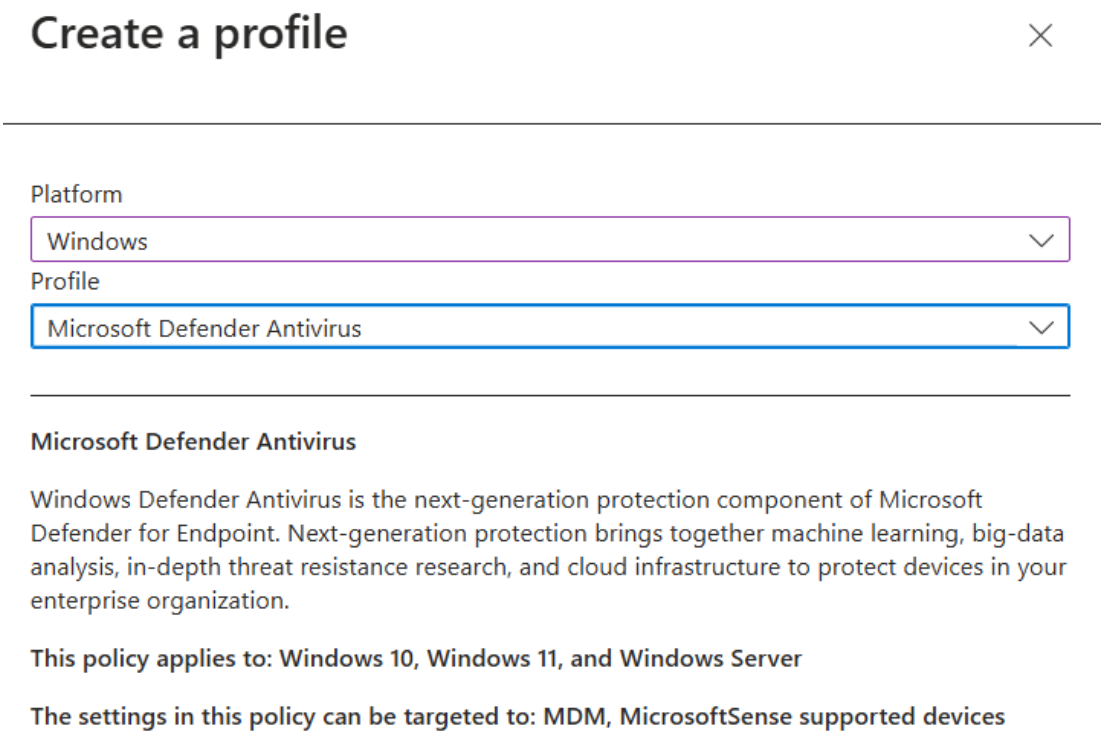
- Endpoint Detection and Response
- Create a new EDR policy targeting Windows. Target all devices.



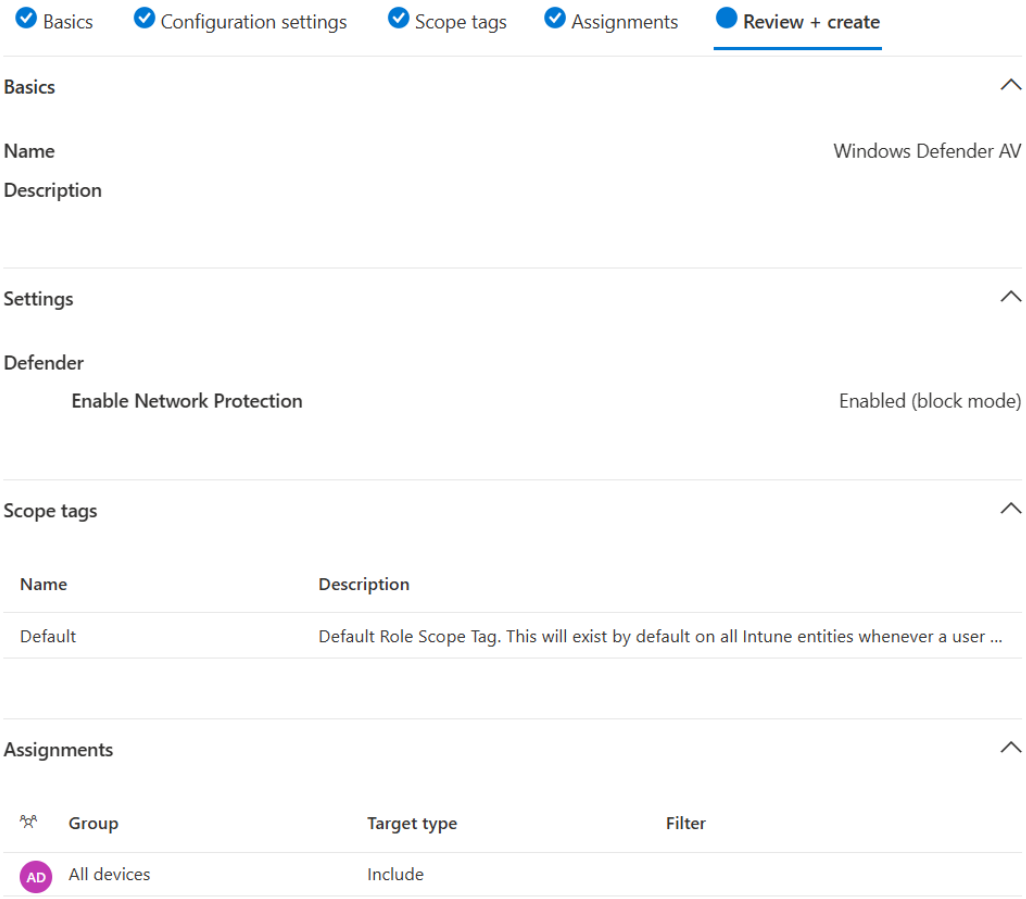


- Antivirus

- Create a new Microsoft Defender Antivirus profile



- Enable Network Protection in Block mode. Target all devices.

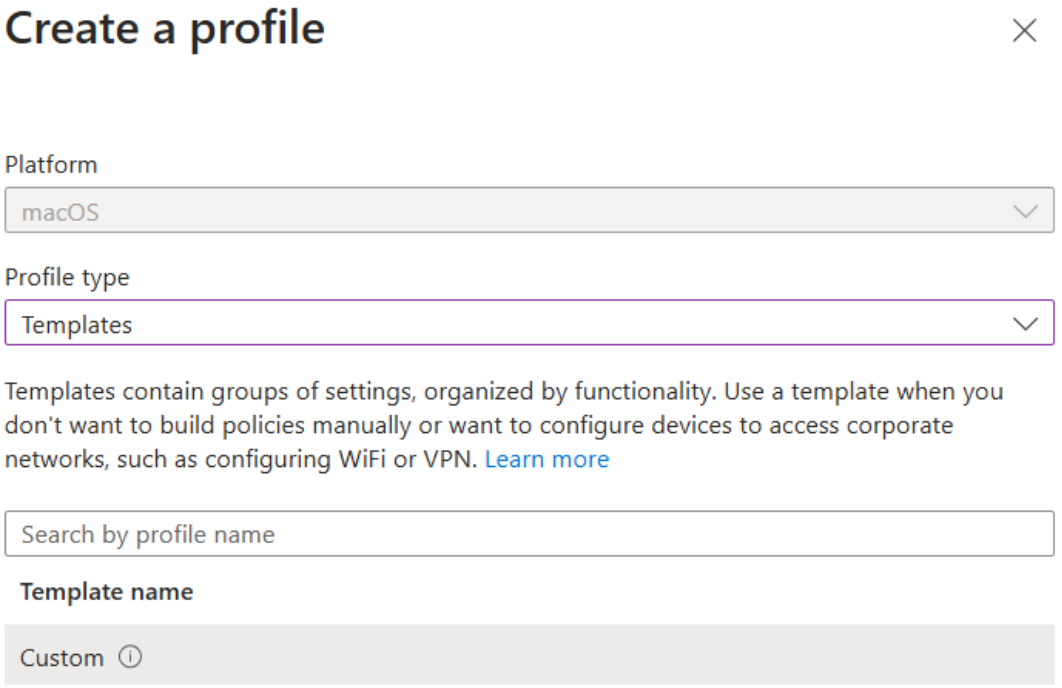


MacOS

Deploying MDE on MacOS is a multi-step manual process, and changes occasionally. Refer to Intune-based deployment for Microsoft Defender for Endpoint on Mac - Microsoft Defender for Endpoint | Microsoft Learn for the most current steps.

tl;dr
If you want a sample combined deployment, I've combined mobileconfig files here to set the following settings

- AutoUpdate enabled, broad channel
- Network protection set to block
- All other required mobileconfig settings, such as full disk access, etc.
- Deploy the combined profile
- Create a device configuration profile for macOS devices using a custom template



- For configuration settings, upload the mobileconfig from above. Target device channel.
- Target all MacOS devices
- Deploy MDE
- Deploy the MDE packageMDE App in Intune
- Deploy the Onboarding Package

- Download the MDM/Intune onboarding package from Defender XDR

Select operating system to start onboarding process:

macOS

Onboard macOS devices through Microsoft Defender

1. Install the agent and onboard a device

Connectivity type

Streamlined

This package allows devices to onboard using streamlined connectivity method. Check devices you onboard meet specific prerequisites. Install the agent on the macOS device using the installation package, then onboard devices to Microsoft Defender using the configuration package that matches your preferred deployment method. For other device preparation instructions, read Onboard and set up.

Note: Streamlined connectivity is supported on MDE product version released in February 2024 and above.

Deployment method

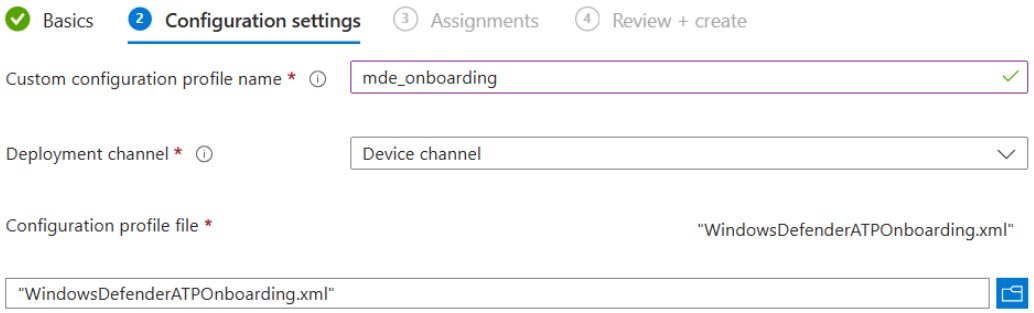
Mobile Device Management / Microsoft ...

You can use Mobile Device Management solutions, such as Microsoft Intune to configure and monitor your devices.
Before downloading the packages, review the instructions.

Download installation package

Download onboarding package

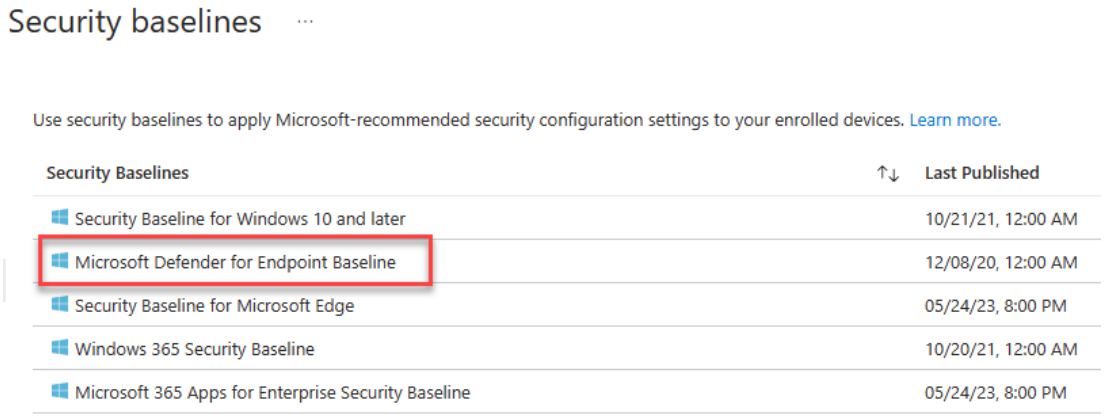
- Deploy via Intune as a Custom Config template




```xml
<?xml version="1.0" encoding="utf-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/Property
<plist version="1">
    <dict>
        <key>PayloadUUID</key>
        <string>A27F524F-7A54-4E9A-B459-B50A321C4295</string>
        <key>PayloadType</key>
        <string>Configuration</string>
        <key>PayloadOrganization</key>
        <string>Microsoft</string>
        <key>PayloadIdentifier</key>
        <string>A27F524F-7A54-4E9A-B459-B50A321C4295</string>
        <key>PayloadDisplayName</key>
        <string>WDATP settings</string>
        <key>PayloadDescription</key>
        <string>WDATP configuration settings.</string>
        <key>PayloadVersion</key>
        <integer>1</integer>
        <key>PayloadEnabled</key>
        <true/>
```

Security Baselines

- Create a new Microsoft Defender for Endpoint Baseline policy and target all devices.

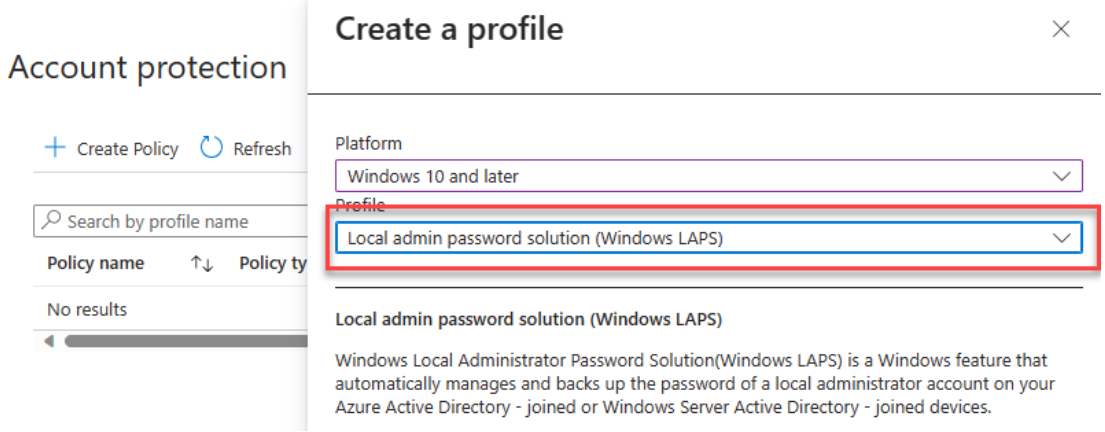


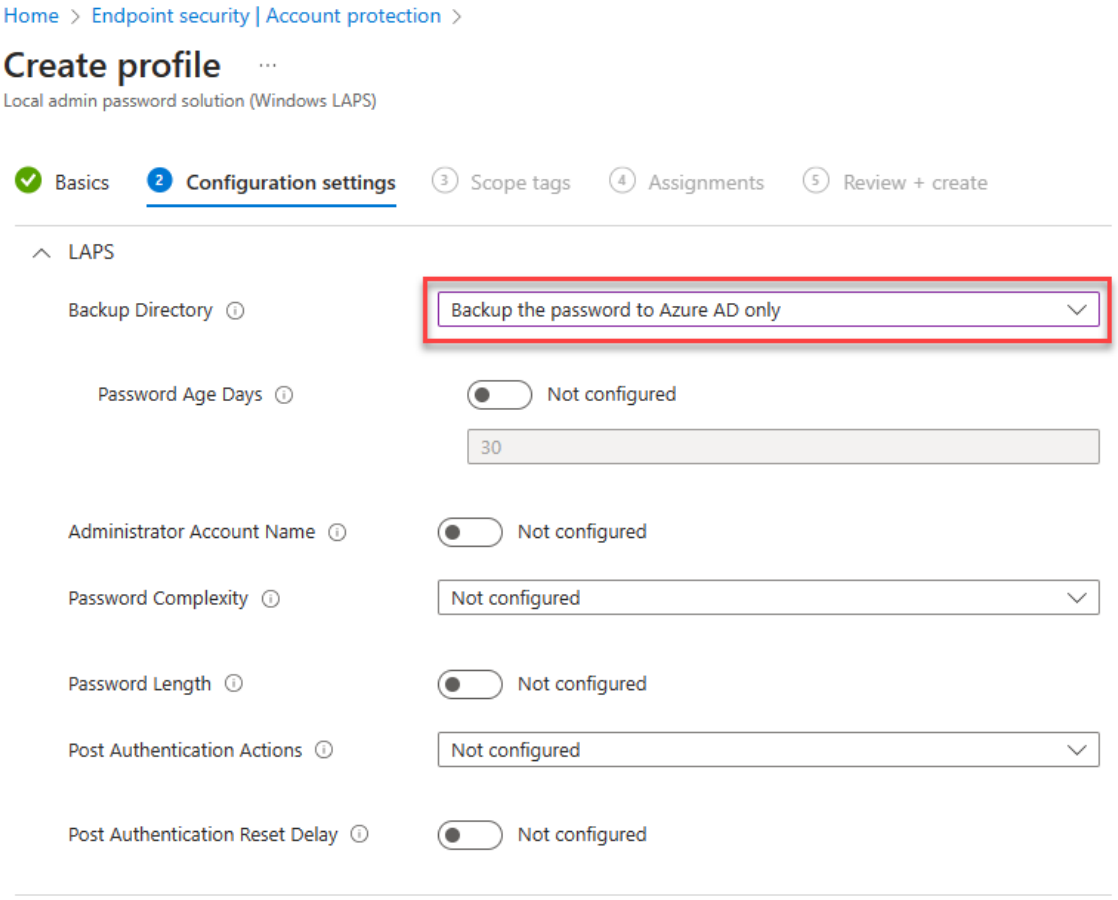
Account Protection (LAPS)

🔗 Portal

- Enable LAPS in the portal

- Create a Windows LAPS profile and apply to all devices.



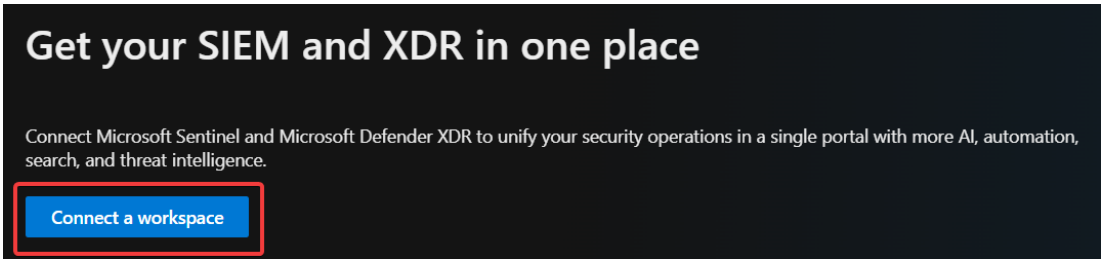


M365 Defender

🔗 Portal

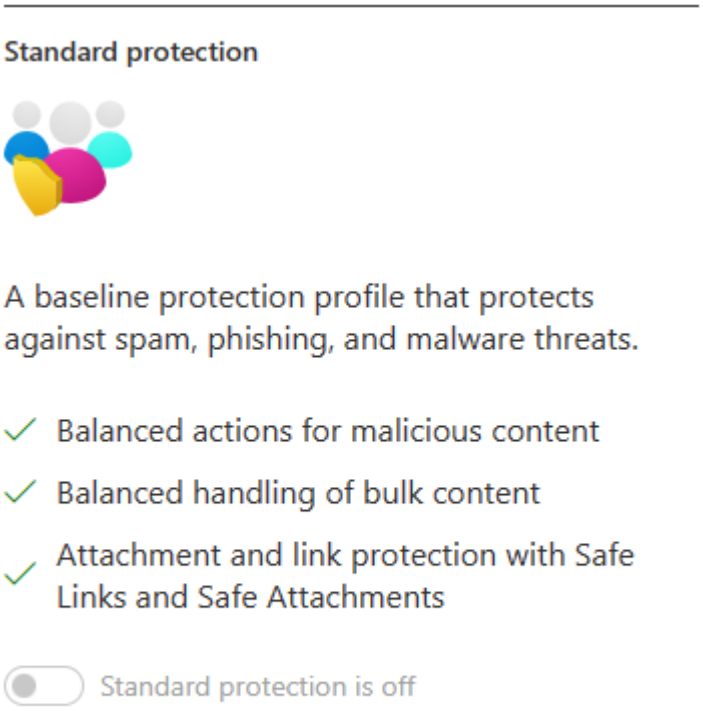
XDR

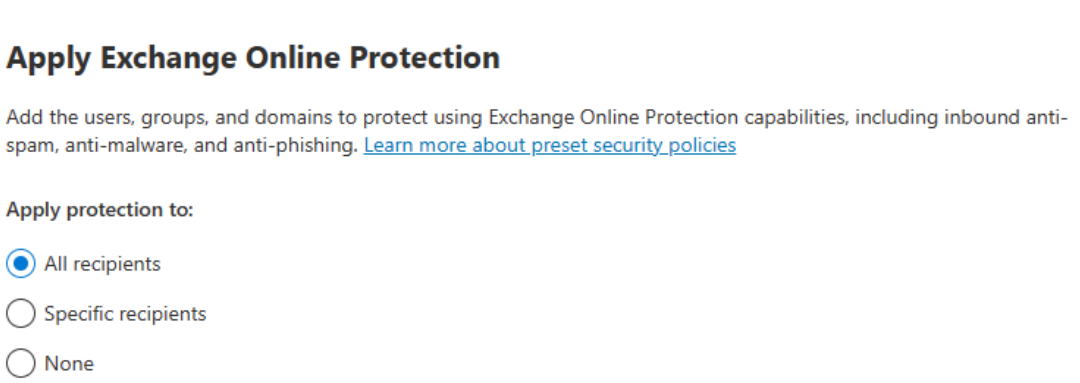
- Enable unified SIEM and XDR.



Email & collaboration

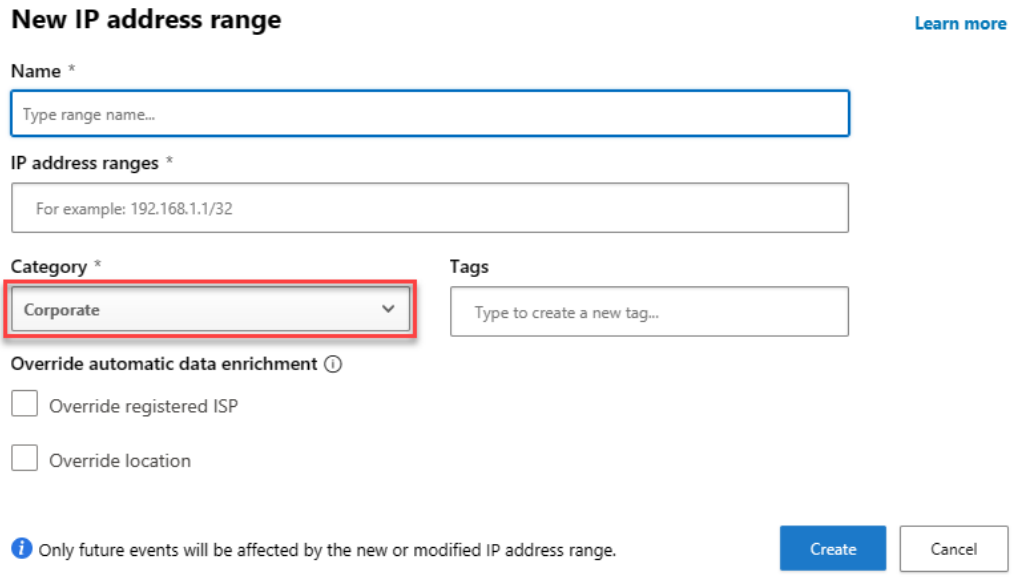
- Peset Security Configuration Policies
- Enable Standard Protection Preset Policies.



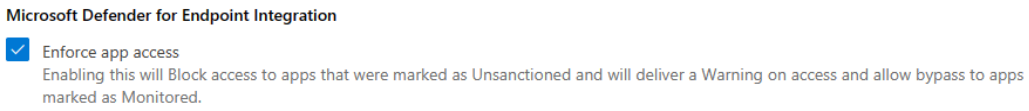


MDCA

- System
- IP Address Ranges
  - If you have IP Ranges as Trusted Named Locations in EID, add them as Custom IP Address Ranges in MDCA with the category of Corporate



- Cloud Discovery
- Defender for Endpoint
  - Enforce App Access with Defender for Endpoint



- User Enrichment

- Enable User Enrichment

  User enrichment

  User enrichment automatically matches, enriches and replaces discovered user identifiers with their Azure Active Directory username.

  ☑ Enrich discovered user identifiers with Azure Active Directory usernames.
  Enabling this after you uploaded data might result in user duplication.

- Information Protection
- Microsoft Information Protection
  - Enable automatically scan new files
  - Enable scanning protected files. You'll need to go through the OAUTH grant process.

  Microsoft Information Protection settings

  ☑ Automatically scan new files for Microsoft Information Protection sensitivity labels and content inspection warnings
  When enabled, the App connector will scan new files, searching for sensitivity labels embedded by Microsoft Information Protection.

  ☐ Only scan files for Microsoft Information Protection sensitivity labels and content inspection warnings from this tenant
  When enabled, all Microsoft Information Protection sensitivity labels that were set by external tenants will be disregarded.

  Get more info in the Microsoft Information Protection integration guide.

  Save We secure your data as described in our privacy statement and online service terms .

  Inspect protected files

  File policies can inspect content and/or read labels in Microsoft Information Protection protected files.
  To inspect protected files, read labels from protested files, grant Defender for Cloud Apps permission in Azure AD.

  Active
  Protected files can be inspected by file policies. Learn more

- Files
  - Enable file monitoring

  ☑ Enable file monitoring
  This enables to see files in your SaaS apps.

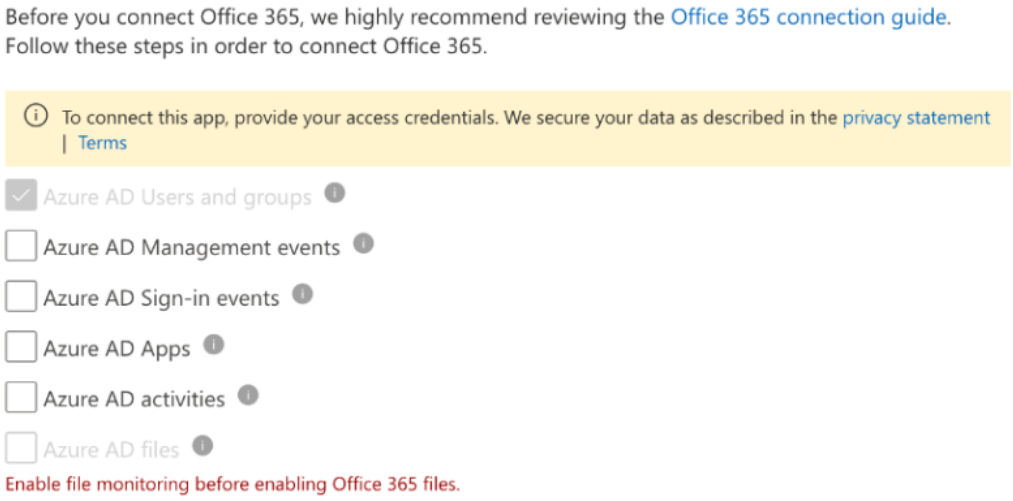
- App governance
- Service Status
  - Turn on app governance

  Get comprehensive visibility and control over cloud apps that authenticate through Azure Active Directory, Google, and Salesforce. Learn more about app governance

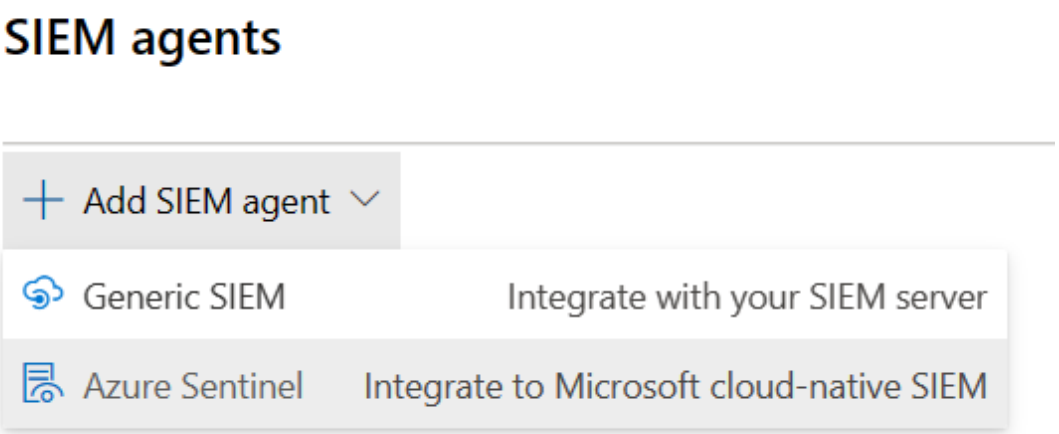
  ○ Use app governance

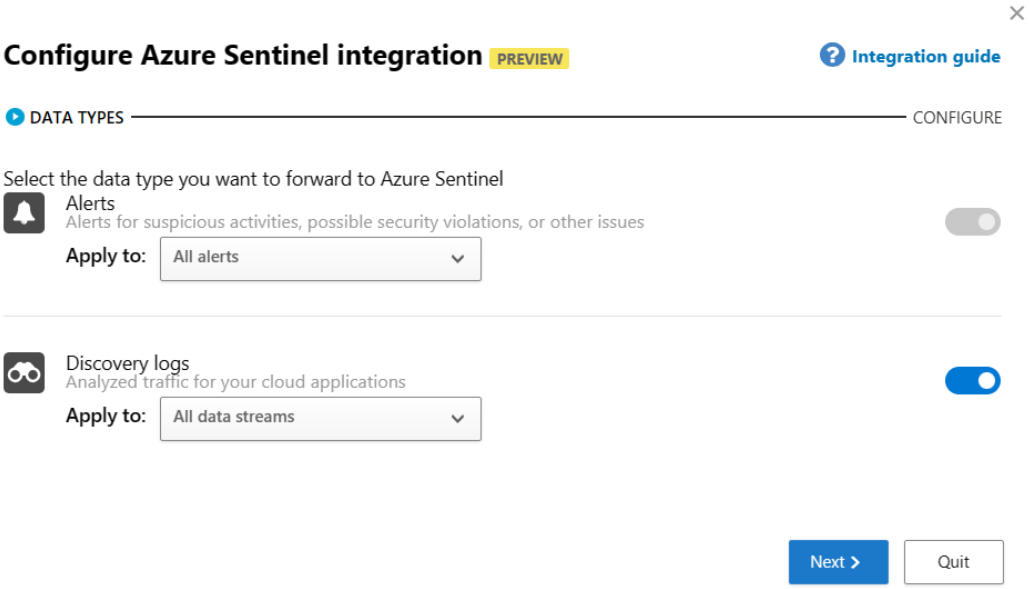
- Connected Apps
- App Connectors

- Click Connect an app, choose Microsoft 365 from the list. Select all options.



- SIEM Agents
- Add the Azure Sentinel integration





Endpoints

- Advanced Features

- Ensure your settings match those below:

This section provides a set of advanced features you can enable. These features require integration with other products. You need to verify that these settings are enabled to use the features.

Off

Restrict correlation to within scoped device groups

When this setting is turned on, alerts are correlated into separate incidents based on their scoped device group. By default, incident correlation happens across the entire tenant scope.

ⓘ Changing this setting impacts future alert correlations only.

Off

Enable EDR in block mode

When turned on, Microsoft Defender for Endpoint leverages behavioral blocking and containment capabilities by blocking malicious artifacts or behaviors observed through post-breach endpoint detection and response (EDR) capabilities. This feature does not change how Microsoft Defender for Endpoint performs detection, alert generation, and incident correlation. To get the best protection, make sure to apply security baselines in Intune. See EDR in block mode for more details.

On

Automatically resolve alerts

Resolves an alert if Automated investigation finds no threats or has successfully remediated all malicious artifacts.

On

Allow or block file

Make sure that Windows Defender Antivirus is turned on and the cloud-based protection feature is enabled in your organization to use the allow or block file feature.

On

Hide potential duplicate device records

When turned on, this setting will hide duplications that might occur for the following reasons:

- Devices that were discovered more than once
- Discovery of onboarded devices
- Unintentionally discovered onboarded devices

These duplications will be hidden from multiple experiences in the portal to create a more accurate view of the device inventory. The affected areas in the portal include the Device Inventory, Microsoft Defender Vulnerability Management screens, and Public API for machines data. Notably, you will still be able to view these devices in global search, advanced hunting and alert and incidents pages.

ⓘ When activated, this heuristic might hide some discovered devices in certain cases. You can always come back here and choose to view all devices.

On

Custom network indicators

Configures devices to allow or block connections to IP addresses, domains, or URLs in your custom indicator lists. To use this feature, devices must be running Windows 10 version 1709 or later. They should also have network protection in block mode and version 4.18.1906.3 or later of the antimalware platform (see KB 4052623). Note that network protection leverages reputation services that process requests in locations that might be outside of the location you have selected for your Microsoft Defender for Endpoint data.

On

Tamper protection

Stop unwanted changes to your security solution and its essential functions. With tamper protection, malicious apps are prevented from turning off security features like virus & threat protection, behavior monitoring, cloud-delivered protection, and more. Learn about tamper protection requirements

On

Show user details

Enables displaying user details: picture, name, title, department, stored in Azure Active Directory.

On

Skype for business integration

Enables 1-click communication with users.

On

Microsoft Defender for Cloud Apps

Forwards Microsoft Defender for Endpoint signals to Defender for Cloud Apps, giving administrators deeper visibility into both sanctioned cloud apps and shadow IT. It also gives them the ability to block unauthorized applications when the custom network indicators setting is turned on. Forwarded data is stored and processed in the same location as your Cloud App Security data. This feature is available with an E5 license for Enterprise Mobility + Security on devices running Windows 10 version 1709 (OS Build 16299.1085 with KB4493441), Windows 10 version 1803 (OS Build 17134.704 with KB4493464), Windows 10 version 1809 (OS Build 17763.379 with KB4489899) or later Windows 10 versions.

On

Web content filtering

Block access to websites containing unwanted content and track web activity across all domains. To specify the web content categories you want to block, create a web content filtering policy. Ensure you have network protection in block mode when deploying the Microsoft Defender for Endpoint security baseline.

On

Unified audit log

When an audited activity is performed by a user or admin, an audit record is generated and stored in the Office 365 audit log for your organization. For more information, see the Search the audit log in the Security & Compliance Center.

On

Device discovery

Allows onboarded devices to discover unmanaged devices in your network and assess vulnerabilities and risks. For more information, see Device discovery settings to configure discovery settings.

On

Download quarantined files

Backup quarantined files in a secure and compliant location so they can be downloaded directly from quarantine.

On

Default to streamlined connectivity when onboarding devices in Defender portal

With streamlined connectivity, devices connect to fewer URLs and static IPs. You'll still be able to select standard connectivity. Learn about streamlined connectivity

To avoid service connectivity issues, update devices and ensure they can connect to '*.endpoint.security.microsoft.com' before onboarding. View requirements

On

Apply streamlined connectivity settings to devices managed by Intune and Defender for Cloud

With streamlined connectivity, devices connect to fewer URLs and static IPs. Learn about streamlined connectivity

These settings will apply with

- new EDR policies that select "Auto from connector" in Intune and
- new devices added to Defender for Cloud.

To avoid connectivity issues, update devices and ensure they can connect to "*.endpoint.security.microsoft.com" before turning this feature on. View requirements

On

Live Response

Allows users with appropriate RBAC permissions to investigate devices that they are authorized to access, using a remote shell connection.

On

Live Response for Servers

Allows users with Live Response privileges to connect remotely to servers (Windows Server or Linux devices) that they are authorized to access.

Off

Live Response unsigned script execution

Enables using unsigned PowerShell scripts in Live Response.

On

Deception

Manage and deploy lures and decoys to catch attackers in your environment. After you turn this on, go to Rules > Deception rules to run deception campaigns.

On

Share endpoint alerts with Microsoft Compliance Center

Forwards endpoint security alerts and their triage status to Microsoft Compliance Center, allowing you to enhance insider risk management policies with alerts and remediate internal risks before they cause harm. Forwarded data is processed and stored in the same location as your Office 365 data.

On

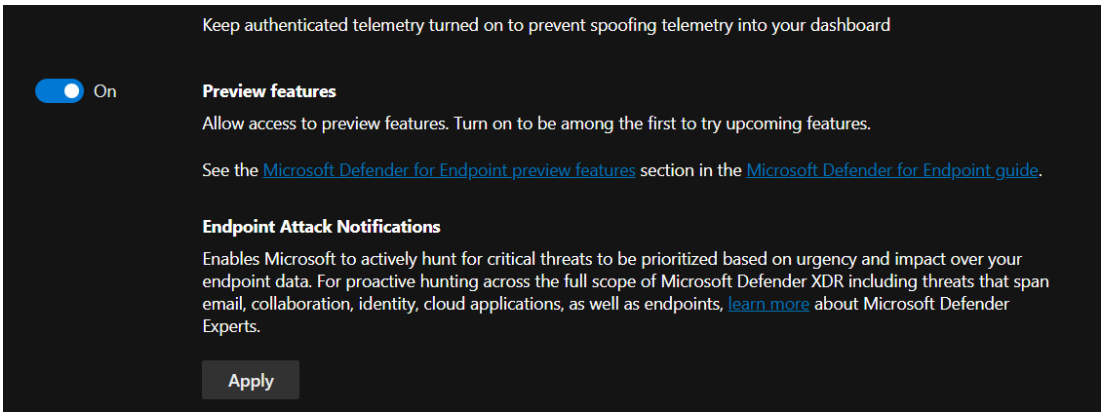
Microsoft Intune connection

Connects to Microsoft Intune to enable sharing of device information and enhanced policy enforcement.

Intune provides additional information about managed devices for secure score. It can use risk information to enforce conditional access and other security policies.

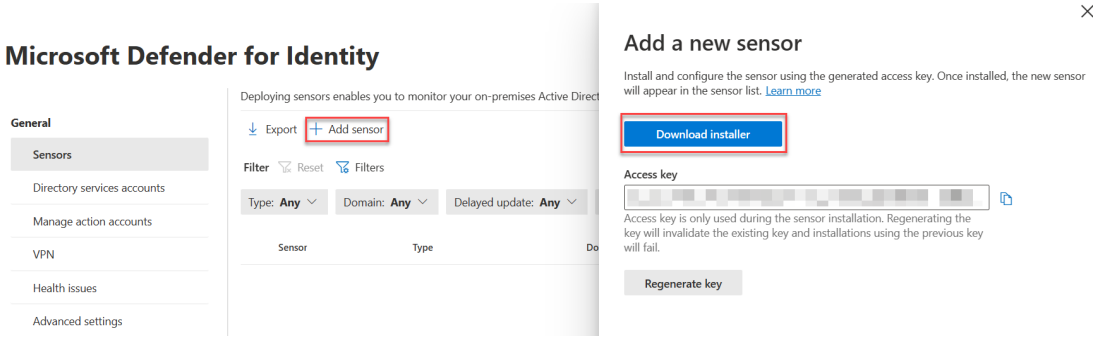
On

Authenticated telemetry

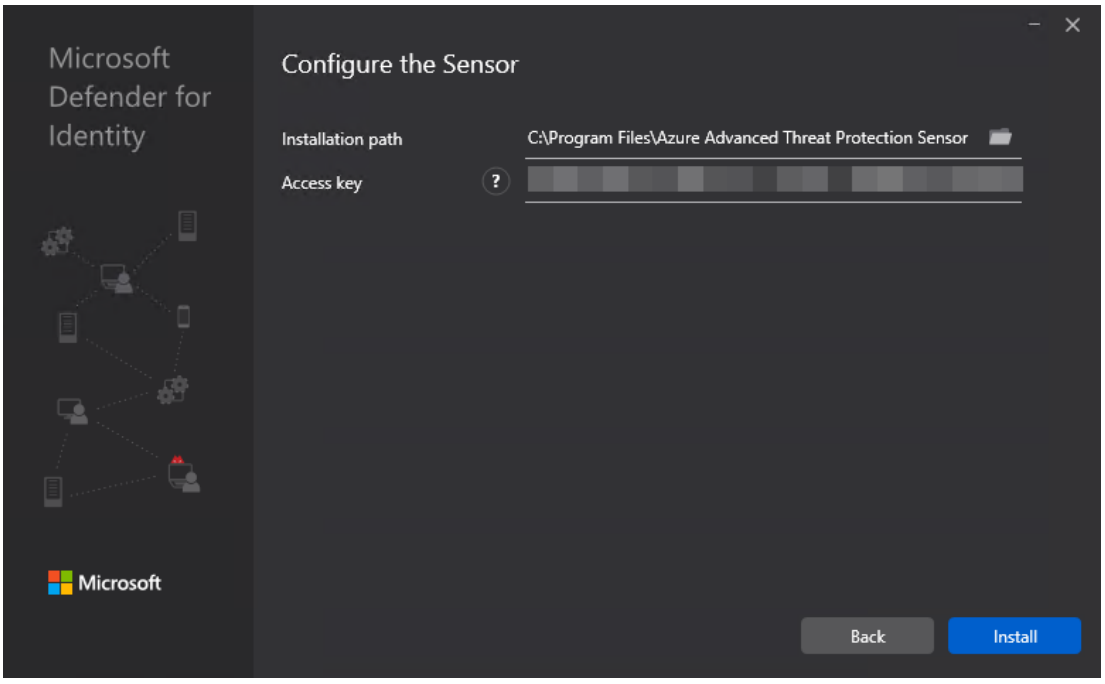


Identities

- Sensors
- Click +Add Sensor, and download the installer and copy the Access key



- Install the sensor on all DCs in AD. Use the access key when prompted by the installer.



- Active Directory
- Configure Event Collection via GPO
- Configure Group Managed Service Account account

- On the first DC
- Create root KDS key

```
Add-KdsRootKey -EffectiveTime ((get-date).addhours(-10))
```

- Purge kerberos tickets

```
klist purge -li 0x3e7
```

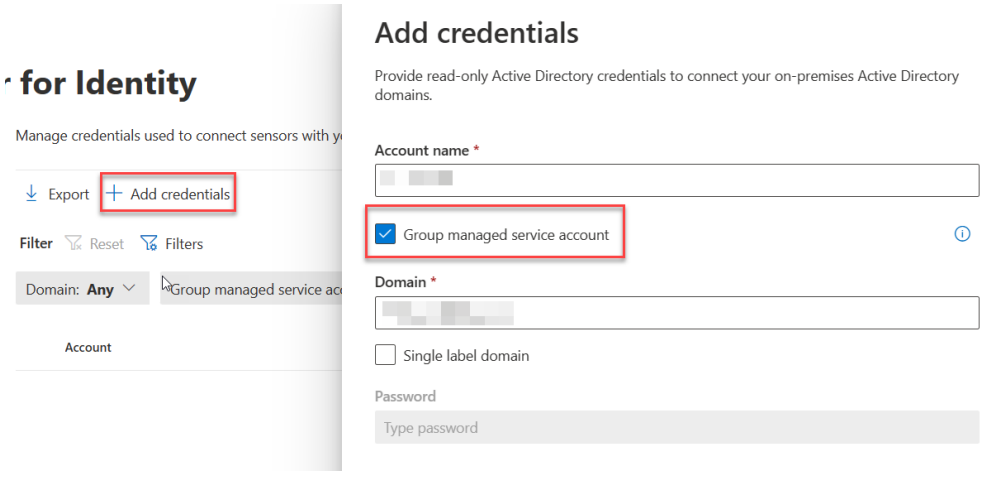
- Create the gMSA

```
New-ADServiceAccount accountname -
PrincipalsAllowedToRetrieveManagedPassword "Domain Controllers" -
DNSHostName accountname.domain.contoso.com
```

- Install the gMSA on the DC

```
Install-ADServiceAccount -Identity 'accountname'
```

- On the other DCs, purge kerberos tickets and install the service account
  - Add the gMSA in the portal



Purview

Advanced Audit

🔗 Portal

- Enable Auditing
- With the ExchangeOnlineManagement module in PS5/PS7+

```
Enable-OrganizationCustomization
Set-AdminAuditLogConfig -UnifiedAuditLogIngestionEnabled $true
```

Device Onboarding

🔗 Portal 📖 Windows 📖 Mac

- Enable Windows and Mac device onboarding. This requires MDE.

Sensitivity Labels

Enable labeling for Protected content & PDFs

📖 Enable Sensitivity Labels for Protected Content 📖 Enable labeling support for PDFs

- Enable labeling for Protected content & PDFs
- Using the SharePoint Module in PowerShell 5

```
connect-sposervice -url 'https://<tenant>-admin.sharepoint.com/'
Set-SPOTenant -EnableAIPIntegration $true
Set-SPOTenant -EnableSensitivityLabelforPDF $true
```

Enable Labeling for Containers

In a fresh tenant, there will not be any EntraID group settings configured, so those need to be created. Then enable labeling for containers. After that, you can enable the label sync.

- With the Graph SDK in PS7+

```
Connect-MgGraph -Scopes "Directory.ReadWrite.All"
$TemplateId = (Get-MgBetaDirectorySettingTemplate | where { $_.DisplayName
-eq "Group.Unified" }).Id
$params = @{
   templateId = "$TemplateId"
   values = @(
      @{
         name = "EnableMIPLabels"
         value = "True"
      }
   )
}
New-MgBetaDirectorySetting -BodyParameter $params
```

- With the ExchangeOnlineManagement module in PS5/PS7+

```
Connect-IPPSSession
Execute-AzureAdLabelSync
```

Enable co-authoring for Encrypted Files

This can be done in the portal, or via PowerShell.

- With the ExchangeOnlineManagement module in PS5/PS7+

```
Connect-IPPSSession
Set-PolicyConfig -EnableLabelCoauth:$true
```

DLP

Endpoint DLP

SETTINGS

🔗 Portal 📘 Docs

In Settings, change the following:

- Advanced classification scanning and protection: `On`

BROWSER EXTENSION

MIP Scanner

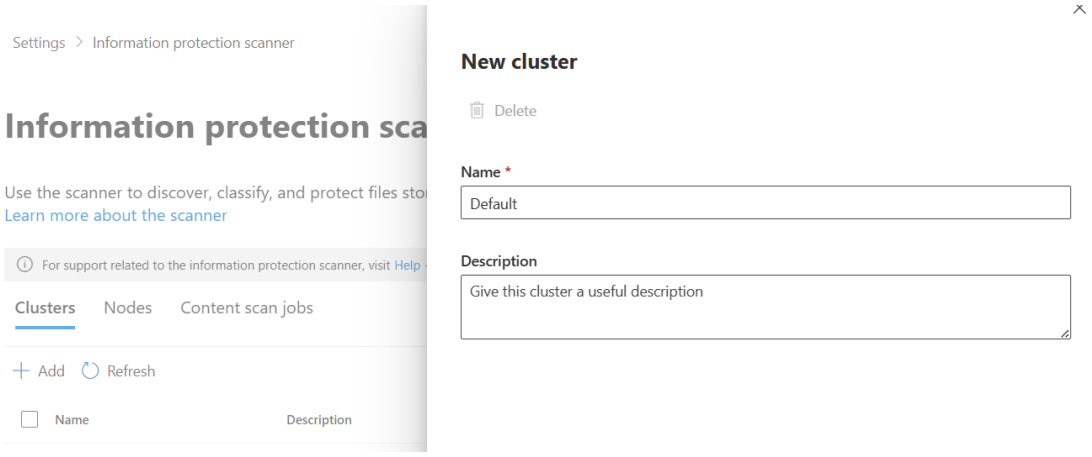
🔗 Portal 📘 Docs

Prerequisite

- Service account in AD, exclude from MFA registration and CAs
- SQL server for the scanner, as well as a windows server.

Deployment

- Create a Scanner Cluster



- Create a Content Scan Job. Be sure to disable any of the auto options - this will just be for scanning.

Edit content scan job

General

Content scan jobs specify how the on-premise repositories should be scanned

Content scan job name *

Default

Description

Cluster

Default

Schedule

Manual

Info types to be discovered

All

Treat recommended labeling as automatic

Off

Enable DLP policy rules

Off

Enforce sensitivity labeling policy

Off

Label files based on content

On

Default label

None

Relabel files

Off

Preserve "Date modified", "Last modified", and "Modified by"

On

Include or exclude file type to scan *

Exclude

Include

Include or exclude file type to scan

.lnk,.exe,.com,.cmd,.bat,.dll,.ini,.pst,.sca,.drm,.sys,.cpl,.inf,.drv,.dat,.tmp,.msp,.msi,.pdb,.jar,.oc...

Default owner

Scanner Account

Set repository owner

Off

Exact Data Match

🔗 Portal 📘 Docs

EDM SIT

- Create a new EDM SIT. Since I work in healthcare, I typically use Synthea to generate patient records. We do have sample industry files you can use.
- Make note of the datastrore name when you finish the EDM wizard. You'll need it for the EDM uploader.

EDM UPLOADER TOOL

- Create a service account for the EDM upload agents to run as.
- Create a EntraID security group named `EDM_DataUploaders` and add the service account to it.
- Install the EDM upload tool to `c:\EDM`
- Place sample data in `c:\EDM\Data`
- Save the schema `.\EdmUploadAgent.exe /SaveSchema /DataStoreName your_data_store_name /OutputDir c:\edm\data`
- Create `c:\EDM\hash`
- Upload the data `.\EdmUploadAgent.exe /uploaddata /datastorename your_data_store_name /datafile C:\edm\data\your_data.csv /hashlocation c:\edm\hash /schema C:\edm\data\your_data_store_name.xml /allowedbadlinespercentage 5`

Insider Risk Management

Roles

🔗 Portal 📘 Docs

Add your account to the `Insider Risk Management` role.

Browser Activity Plugins

1. Deploy the Edge profile via Intune as described here
2. Deploy the Chrome profile via Intune as described here

Settings

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- Analytics
- Toggle analytics on
- Data sharing
- Enable share data with Defender XDR
- Policy indicators
- Select all indicators under the following categories
  - Office
  - Microsoft Defender for Cloud Apps
  - Cumulative exfiltration detection
  - Risk score boosters
  - Generative AI Apps
  - Microsoft Copilot Experiences
  - Microsoft Entra
  - Risk Detection indicators

ANALYTICS

- Toggle analytics on

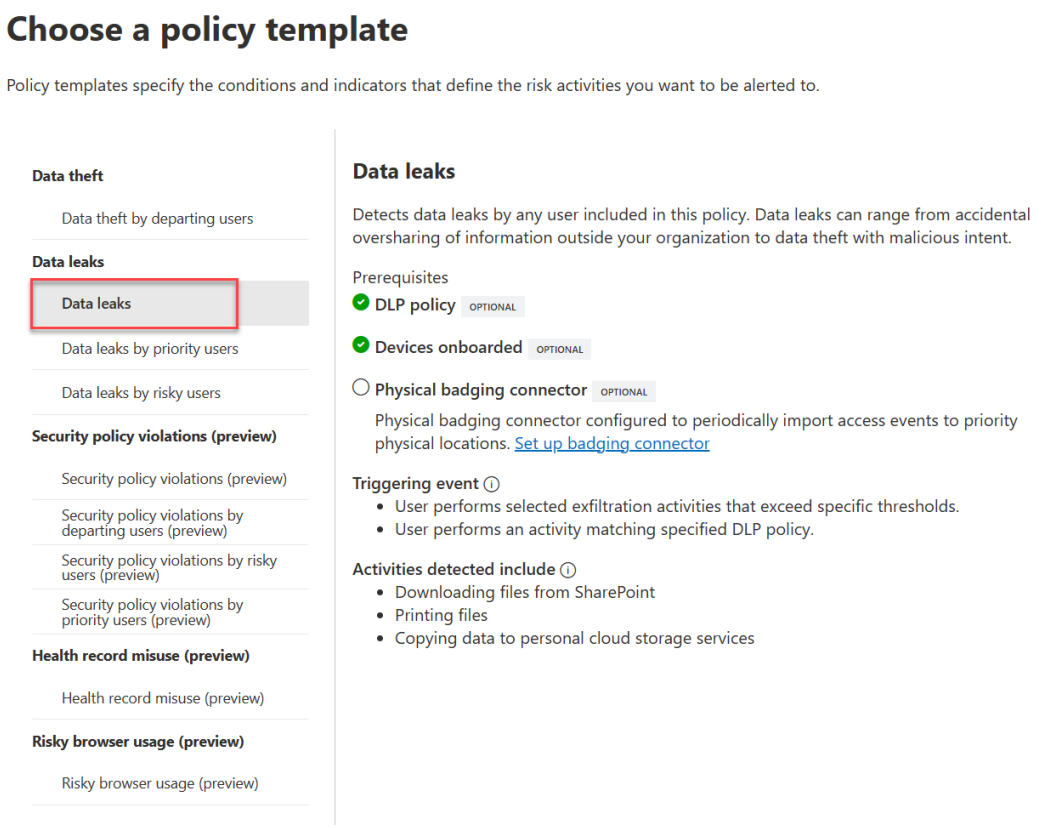
POLICY INDICATORS

- Select all indicators under the following categories
- Office
- Device
- Microsoft Defender for Endpoint
- Risky Browsing
- Microsoft Defender for Cloud Apps

Polices

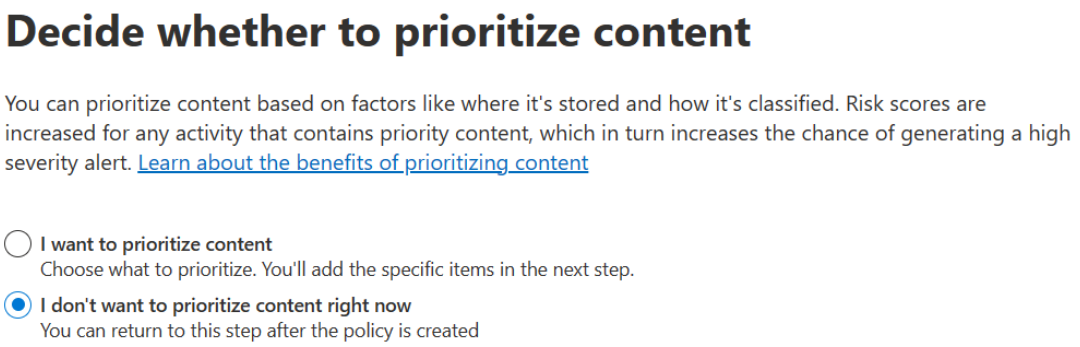
🔗 Portal 📘 Docs

1. Create a Data Leaks policy from the template



2. Target all users

3. Choose not to prioritize content.



4. For triggering events, choose `User performs an exfiltration activity`

5. For thresholds, choose `Apply built-in thresholds.`

6. For indicators, leave the default ones checked.